XK-335B

Mechatronics Automation Production Line Training Device

Experiment Instruction

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Chapter One Project Start

Task Objectives

1. Getting to know the functions, roles and features of the Automatic production Line
2. Briefing on the development of the Automatic Production Line
3. Getting to know the basic structure of XK-335B Automatic Production Line

Task One Getting to know APL and its Application



The APL of plastic case circuit breaker applied in ZhengTai Electric Appliance Co. Ltd. It includes automatic feeding, automatic riveting, 5 times power-on check, transient features' check, time delay features' check and automatic marking. It is controlled by PLC. Each unit has its own control and sounding and lighting alarm function. A perfect network system for the production line is built up by adopting a network technique, upgrading greatly the labor efficiency and product quality.



The picture shows an automatic assembly line of automobile brake in an auto parts factory. Considering the factors of equipment performance, production cycling, general layout and material transportation, etc. it adopts standardized and modularized designs, various manipulators and programmable automation devices in order to achieve process automation in feeding, assembling, detecting, marking and packing. It also adopts network communication monitoring and data process realizes control and management.

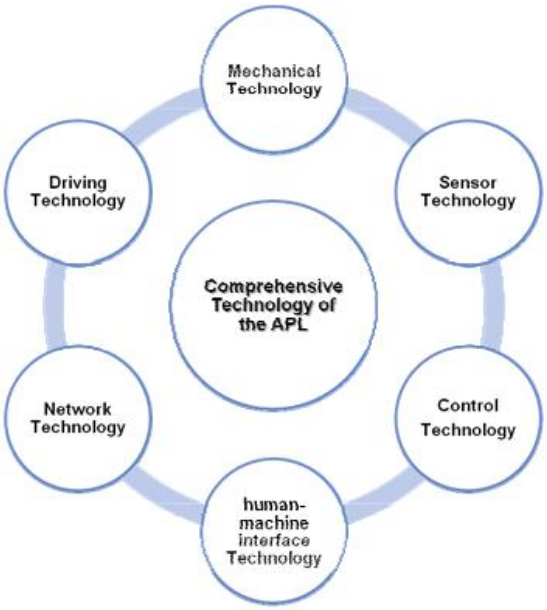


The picture shows an automatic bottling line in a daily-use chemicals factory, which mainly performs production processes of feeding, can filling, sealing, detecting, marking, packing and stacking and fulfills the requirement for mass production.

1. What is the APL?

The APL has gradually developed from a production line. It not only requires each mechanical unit in the line to automatically complete planned processes to make quality products, but also requires all the processes to be performed automatically in work piece loading and unloading, positioning and clamping, work piece delivering, work piece selecting and even packing. It is designed to automatically operate in accordance with a specified program.This type of integrated electromechanical system working automatically is referred to as automatic production line (APL).

Automatic Production Line Technology



It integrates various machining devices in accordance with technological process with auxiliary devices and associates all the actions by controlling hydraulic pressure, pneumatic pressure and electrical systems.

2. APL Development

Applied PLC

It is a kind of industrial controller that mainly uses sequential control while making loop regulation auxiliary. It does not only have the function of performing logic determination, timing, counting, memorizing and arithmetic operations, hut also be able to control, on a large scale, switching and analog quantities, eliminating disadvantages of complex programming and nonstandard external interfacing, insufficient use of machinery equipment presented while using industrial control computer in the switch control system, resulting n excess functions, higher manufacturing cost and poor adaptability to the project site. The PLC controllers have replaced many sequential controllers, such as relay control logic, for a lot of their advantages, and widely used in the control of automatic lines.

Applied manufacturer & robot

Robot arms are widely used in automatic lines for work piece loading and unloading, positioning and damping, transferring between procedures, removing surplus of material, processing and packing etc. The third-generation intelligent robot currently under development will not only have the abilities of moving and operating, but also have distinguishing abilities in the sense of sight, hearing and touch. Automatic units that have the abilities of judgment, policy decision and languages are being more and more used in the automatic production lines.

Applied sensor

With the development of material science and the advent of effects of solid physics In succession, the sensor technology, a complete and independent scientific system, has been formed and established. An applied "Intelligent Sensor" with micro-processor has come into being, which is designed to monitor various complex procedures automatically controlled in automatic production lines and is playing a very important role.

Applied hydraulic & pneumatic driving

Especially in the case of pneumatic drive technology, which has won the general attention as it uses inexhaustible air as media, and is characterized by instant driving response, quick in motion, easier to make pneumatic parts, lower cost, convenient for centralized supply and long distance transportation. The pneumatic driving technology has become an independent technical field. It has been developing rapidly and get widely used in various industries, especially in automatic production lines.

Applied network

Namely a leap in the applied network technology

Either a bus at the site or an industrial ether network enables each control unit in the automatic line to form a coordinated operating whole.

Task Two Getting to Know XK-335B APL

1. Getting to Know the Basic Structure of XK-335B



The equipment comprises of 5 units installed on aluminum alloy rail training platform: namely a feeding unit, a processing unit, an assembly unit, a delivery (transfer) unit and a sorting unit. At every station a PLC is installed to perform the function of control. Interconnection between PLC is made through RS485 serial communication, which forms a distribution-type control system.

The task objectives of XK-335B APL is : To transfer a work piece from the feeding bin of the feeding unit to the processing table of the processing unit. After the piece is processed, it is transferred to the material table. Then the work piece is embedded with small columns in different colors out of the feeding bin. The assembled products are then delivered to the sorting unit to be sorted and output. The sorting station sorts out the work piece according to the material and the color of the work piece.

XK-335B can be used for Trainings as follows:

1）Practical training in Automatic Detection Application

2) Practical training in Pneumatic Technology Application

3) Practical training in PLC programming

4) Practical training in PLC networking configuration

5) Practical training in electric control circuit

6) Practical training in frequency converter application

7) Practical training in motor driving and position control

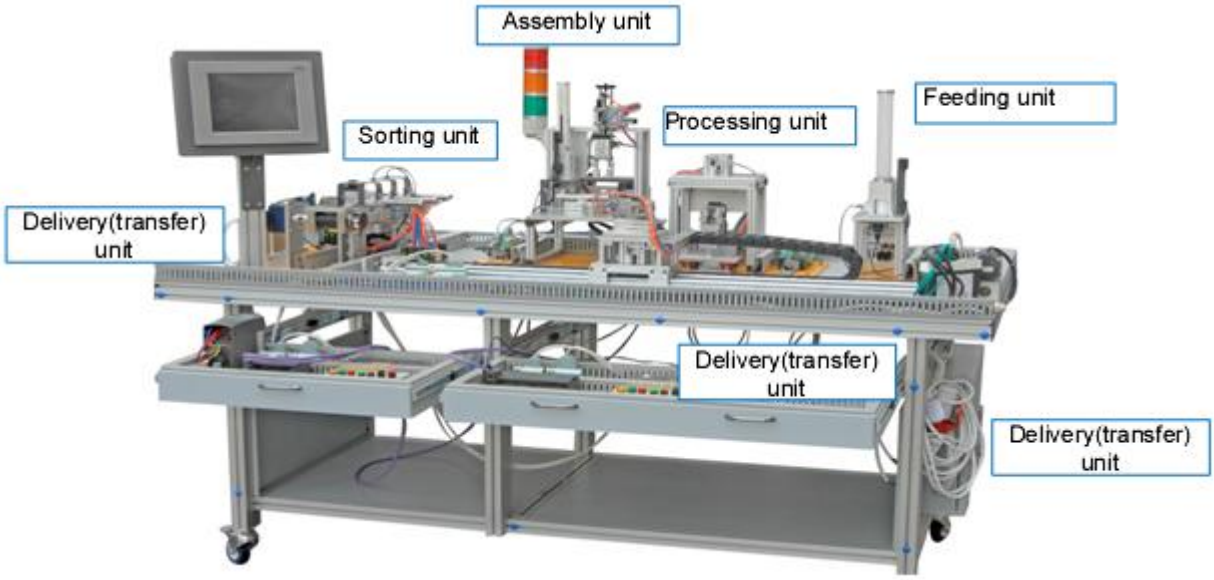
8) Teaching and practical training in automatic control technology

9) Practical training in installing and testing mechanical systems

10) Practical training in system maintenance and fault detection

11) Practical training in configuration programming for touch screen

Overview of XK-335B



2. GETTING TO KNOW THE BASIC STRUCTURE AND FUNCTION OF THE FEEDING UNIT

The feeding unit mainly consists of work piece storehouse, locking devices and pushing-out devices. Main devices include: well-shaped work piece storages, linear cylinders, photoelectric sensors and positioning devices, etc. The basic function of the feeding unit is: automatically push out a work piece to be processed from the feeding bin to the material platform in order to allow manipulator of the delivery unit to pick and take the work piece to other units.

Overview of the Feeding Unit



LEFT VIEW



BACKSIDE VIEW

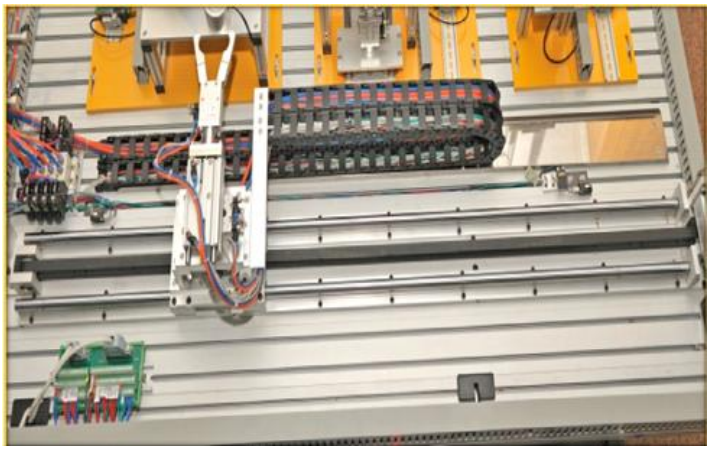
3. Basic Structure and Function of Delivery unit

The delivery unit mainly consists of linear moving devices and work piece picking and delivering

devices. Its main devices include: driving motors, thin cylinders, pneumatic swinging tables, double guide bar cylinders, pneumatic fingers, limit switches and magnetic switches.

Basic function of the delivery unit: It can realize the accurate positioning of the material table to a designated unit. Grasp a work piece and take it to a determined place and drop it.

Diagram of the Robot in Delivery Unit



4. Getting to Know the Basic Structure and Function of the Processing Unit

The processing unit mainly consists of work piece-moving devices and work piece processing devices. Its main devices include: guide rails, linear cylinders, thin cylinders and work piece gripping devices.

Basic functions of the processing unit: deliver a work piece from the material table in the unit (which has been delivered by robot arm of the transfer unit) to the place under the pressing mechanism for once pressing. Then deliver the work piece back to the material table, waiting to be gripped by the manipulator.

Overview of processing unit



5. Getting to know the basic structure and function of the assembly unit

The assembly unit mainly consists of assembly work piece storages and work piece carrying devices. Its main devices include: work piece storages, swinging tables, guide bar cylinders, pneumatic fingers, linear cylinders and photoelectric sensors.

Basic function of assembly unit: Embed work piece of small columns in black or white color from feeding bin into the processed work piece.

Overview of the assembly unit

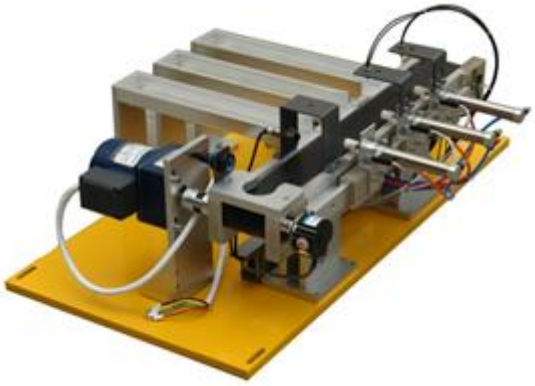


6. Getting to know the basic structure and function of the sorting unit

The sorting unit mainly consists of a belt conveyer and a finished product sorting device. Its main devices include: a linear belt conveyer, a linear cylinder, a three-phase asynchronous motor, a frequency converter, a photoelectric sensor and an optical fiber sensor.

Basic functions of the sorting unit are: Sorting a processed and assembled work piece delivered from the previous unit and allow the work pieces to be carried away through its respective chute according to its color.

Assembly Overview of the Sorting Unit



7．Control system of XK-335B

XK-335B adopts 5PLC of Siemens S7200 series to respectively control 5 units as: delivery unit, feeding unit, processing unit, assembly unit, and sorting unit. PPI Serial bus is adopted for communication among them. Every working unit in XK-335B is controlled by PLC respectively. Each unit can work as an independent system. In addition, they can form a distribution control system by

interconnecting through the network.

When a working unit works as an individual system, its master instruction signals for equipment running and the running status display signal come from the push-button indicator light module of the unit. As shown in figure, the ends of the indicator light and the button on the module are all connected to the terminal block.

XK-335B adopts MCGSTPC series touch screen as its human-machine interface (HMI). Master instruction signals for system operation (reset, startup, and stop) will be displayed through the touch screen HMI during the overall operation. In the meantime, various status information of the system operation will also appear on the HMI. The programming and using of the touch screen will be introduced in the following chapters

Figure of Unit Control Module



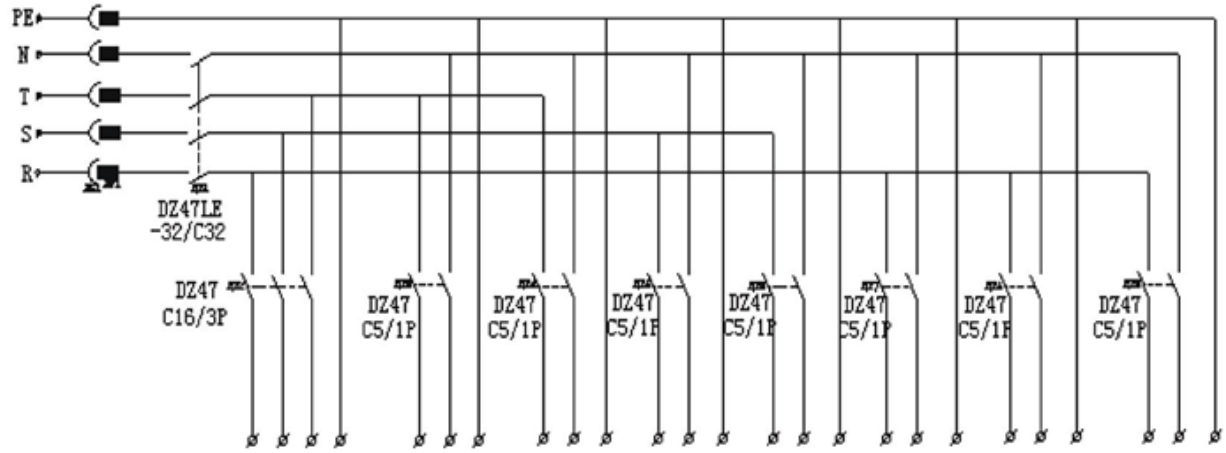
8. Power Supply

The external power supply to XK-335B is of 3-phase & 5-wire AC 380V/220V. Fig. 1-13 shows the primary loop diagram of the power supply, in which the DZ47LE-32/C32 type 3-phase & 4-wire leakage switch is adopted for the main power switch. Each main load in the system is separately supplied with power through an automatic switch, in which the frequency converters are power supplied through DZ47C16/3P three-phrase automatic switches. DZ47C5/2P single phase automatic switches are used for PLC power supply in each working station.

Besides, two DC24V6A voltage regulated power supplies of switch provided in the system are respectively used as direct current supplies to feeding, processing and sorting and delivering units.

Diagram of primary loop for power supply block

3Phase -5wire power supply Incoming line	Main power switch	Frequency converter power control	Servo power control	PLC 1	PLC 2	PLC 3	Switch 1	Switch 2	Spare
				Power supply for feeding, & processing	Power supply for sorting , unit	Power supply for delivery unit	Power DC for feeding, processing & sorting	Power DC for delivery unit	



9. Characteristics of XK-335B

XK-335B is a set of semi-open type equipment. The structure of each working unit is characterized by relative separation of mechanical device from electrical control parts. The mechanical devices of every working unit are integrally installed on base plates. PLC devices which control the operation process for each working unit are installed on drawer plates on each side of the work table. To a certain degree, students may decide the number, and type of units to form equipment as needed. An independent control system can be formed by 5 units at most or by 1 unit only. For a system comprising of more than 1 unit, the PLC network control program can represent the control characteristic of the APL.

Chapter Two Project Preparation

Task One Application of Sensors in the APL

Task Objectives

1. Having a thorough understanding of the structure and characteristics of various sensors for magnetic switches, photoelectric switches, fiber optic photoelectric proximity switches, inductive proximity switches and photoelectric encoders and characteristics of the electrical interface in the production line.

2. Be capable of conducting the installation and testing of various sensors in the Automatic Production Line (APL).

Sensors



Sensors' technology, which is one of the cross, comprehensive and front technologies in the field of measuring technology, semiconductor technology, computer technology, information treatment technology, microelectronics, optics, acoustics, precision mechanism, bionics, materials, is basic foundation of modern technology renovation and information society . It is also indispensible and important part in automatic detection and automatic control technology.

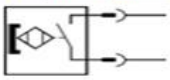
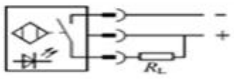
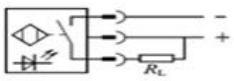
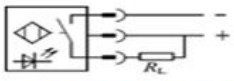
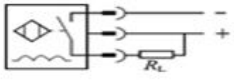


Sensors in the APL are used to sense outer information, detect position, color and they can transfer signals to PLC.



Typical sensors are piezoelectric sensors, magnetic sensors (magnetic switches), photoelectric sensors (photoelectric switches, fiber optic sensors) and so on.

Sensor used in XK-335B

Name of the sensors	Picture of the sensors	Graphics symbol	Use in 'XK-335B
Magnetic switch			Used for the cylinder piston's position detection of each unit in the APL.
Photoelectric switch			Used for the work piece detection in the sorting unit.
Photoelectric switch			Used for the work piece detection in the feeding unit.
Fiber optic sensor		 sharing the photoelectric switch's symbol	Used for the detection for the different color of the work piece in the sorting unit.
Inductive proximity switch			Used for the detection of the different metal workpiece in the sorting unit.
Photoelectric encoder			Used for the driving belt's position control and revolution measuring in the sorting unit

Subtask One Magnetic Switch and Its Application

In XK-335B Automatic Production Line, the magnetic switch is used for the position detection for all types of cylinders. Two magnetic switches are used to detect the positions where the cylinder of the mechanical hand extends and retracts.



Applications of magnetic switches

Magnetic proximity switch real object picture and electrical symbol

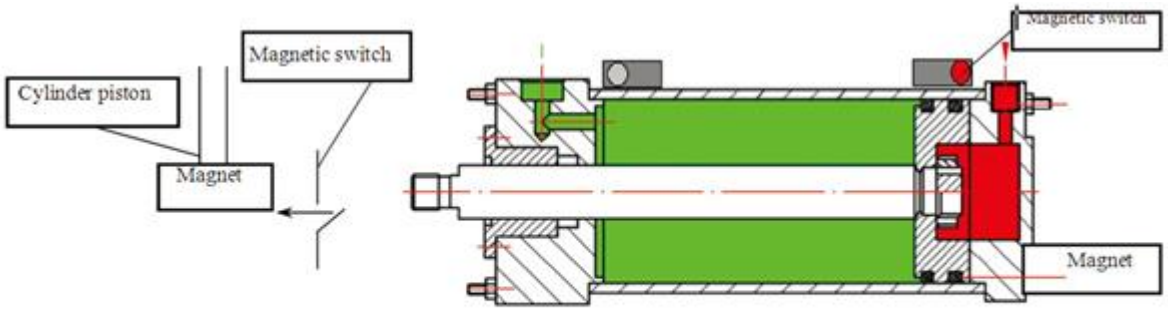


real object picture



electrical symbol

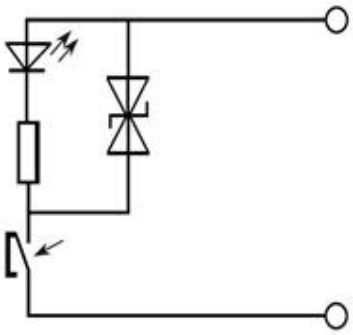
Movement principle of the magnetic proximity switch sensor



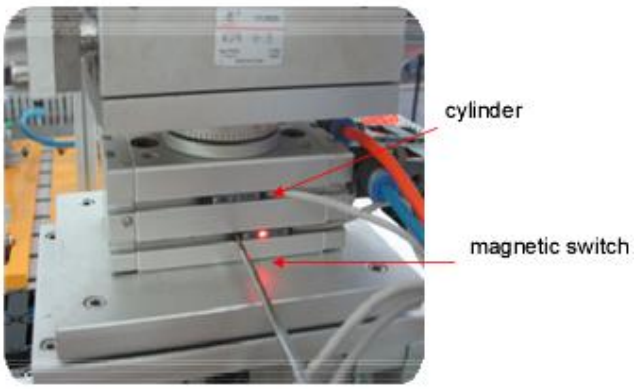
(a) Diagram

(b) Cylinder and magnetic switch

Installation and testing of the magnetic switch's on the cylinder



The Internal circuit of the magnetic switch

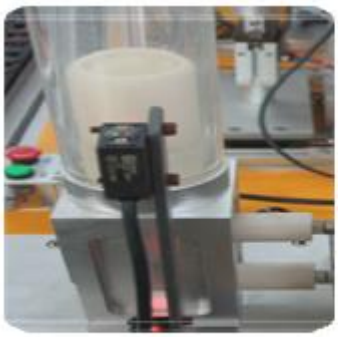


Magnetic switch adjustment

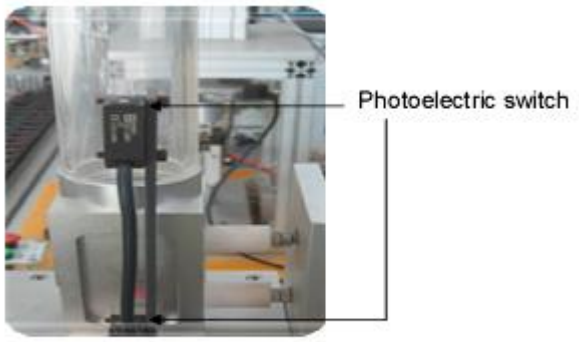
Subtask Two Photoelectric Switch and Its Application

Introduction to photoelectric switches

The photoelectric proximity switch (photoelectric switch for short) is usually used in relatively better and dustless environment. It has no influence on the detected object during its work. Therefore, it is widely used in the production line. In the feeding unit photoelectric switch is used to detect the work piece in the feed bin

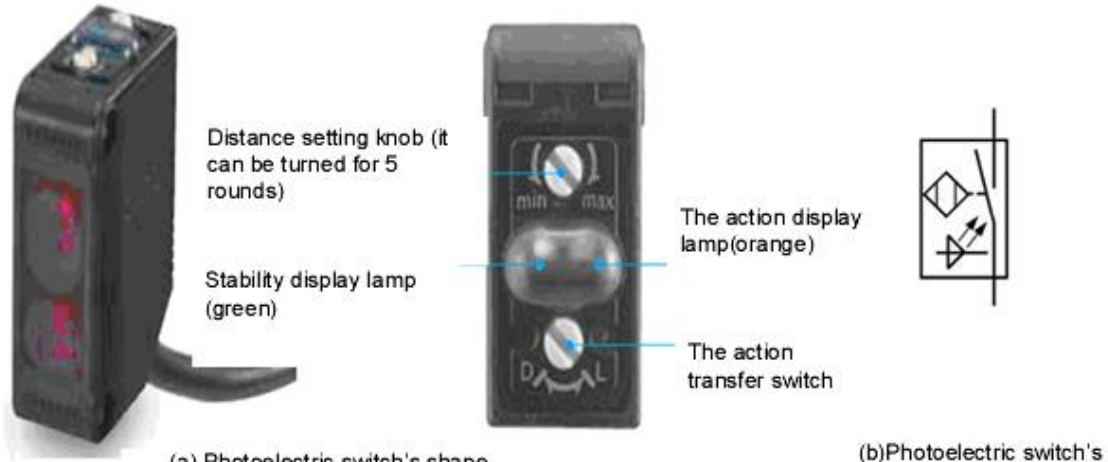


(a) There are work pieces in feed bin



(b) There are no work pieces in feeding bin

Photoelectric switch's shape, adjustment knob, display lamp and electrical symbol

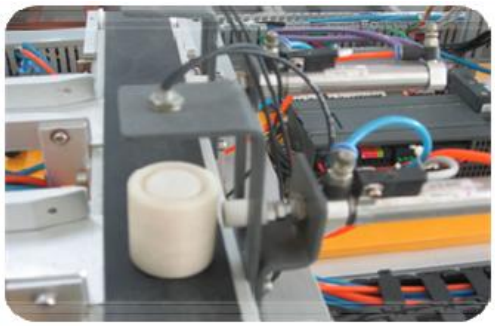


(a) Photoelectric switch's shape

(b)Photoelectric switch's electrical symbol

Subtask Three Fiber Optic Photoelectric Proximity Switch and Its Application

The fiber optic photoelectric proximity switch's application in the sorting unit

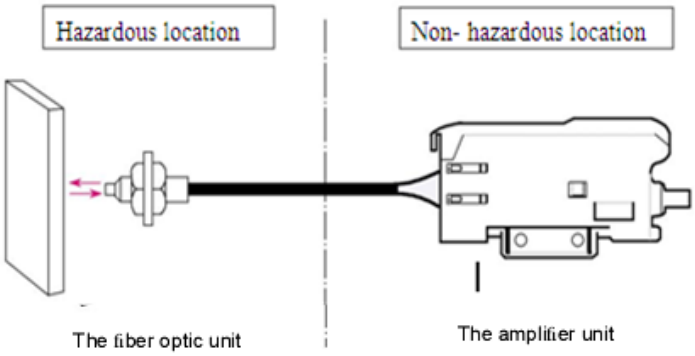


(a) The fiber optic probe



(b) The fiber optic amplifier

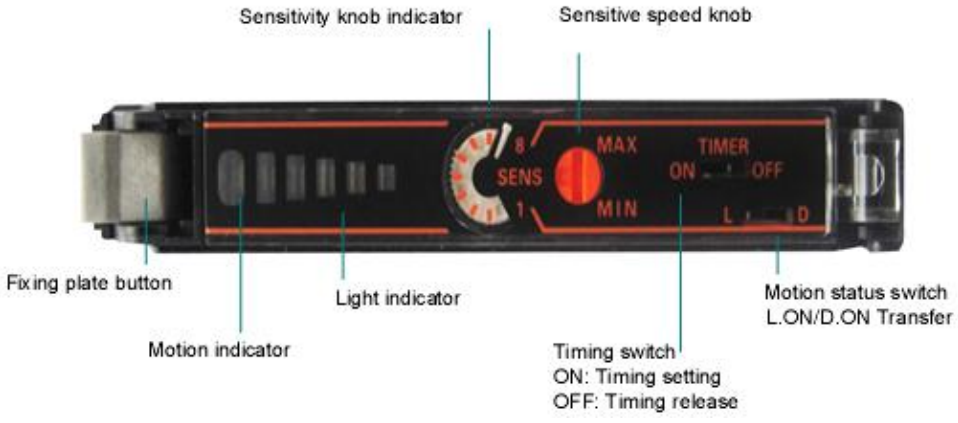
The installation diagram of the fiber optic sensor



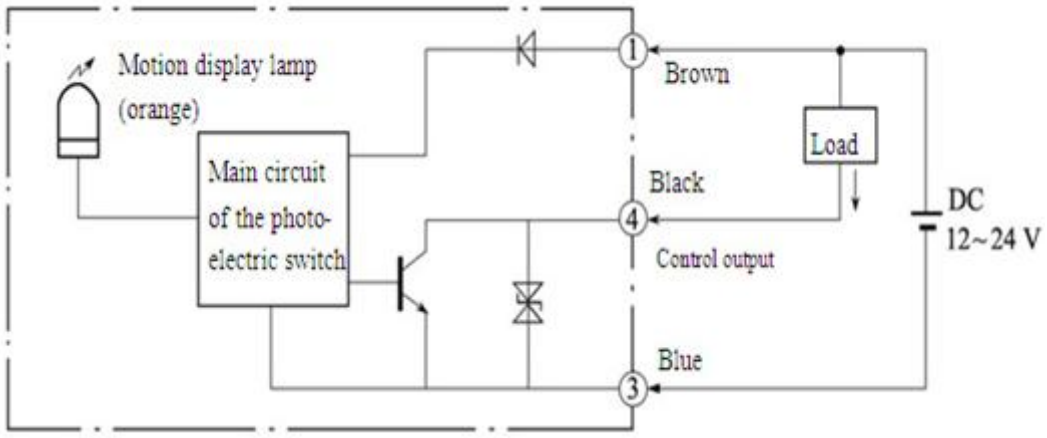
Fiber optic photoelectric switch



Top view of the fiber optic amplifier



The circuit block diagram of the photoelectric sensor



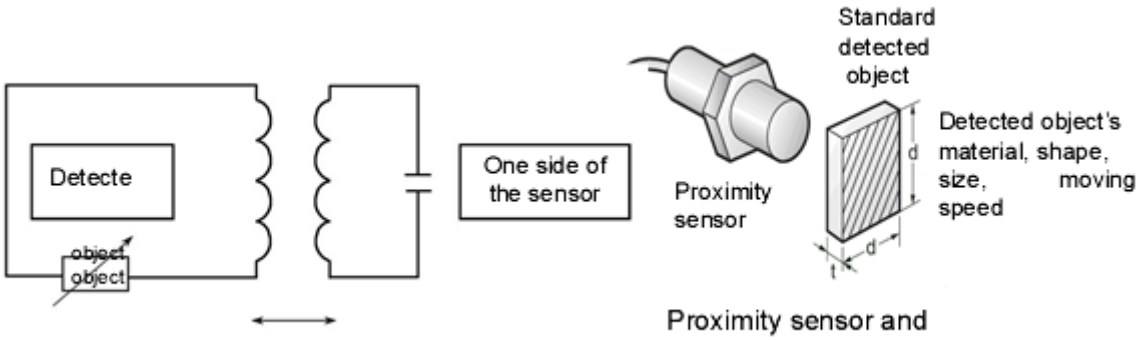
Subtask Four Inductive Proximity Switch and Its Application

In the feeding unit, in order to test whether the work piece to be processed is metallic, an inductive sensor is installed at the base side of the feeding tube



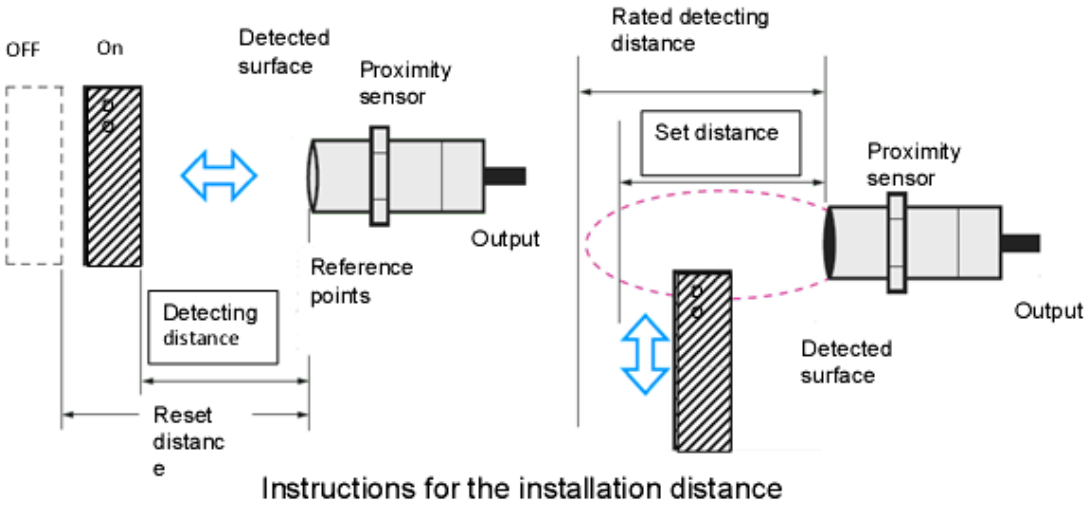
Inductive sensor
in the feeding unit

The electric eddy current inductive proximity switch is a kind of electric inductive sensors. It is a position sensor, which makes use of the eddy current effect.



Proximity sensor and
standard detected object

In the sensor's installation and selection, we should seriously consider the detection distance and set distance to ensure the sensor's reliable movement in the production line.



Instructions for the installation distance

Note:

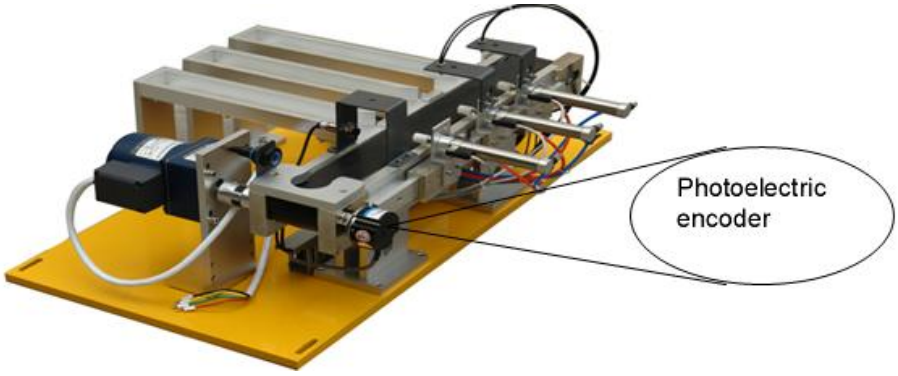
When the situation does not require high accuracy, proximity switches can be used to count the products, measure the revolutions and even the angle of rotation displacement. While in some other situations when a high accuracy is required, the photoelectric encoder can be used to measure the rotation displacement or measure the linear displacement.

Subtask Five Photoelectric Encoder and Its Application

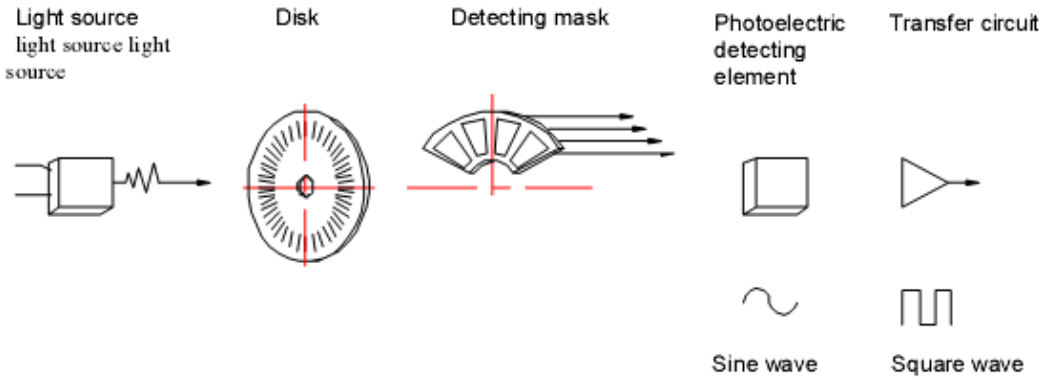
The photoelectric encoder is a kind of sensor that transfers the displacement magnitude of mechanics and geometric into the pulse or digital magnitude through photoelectric conversion.



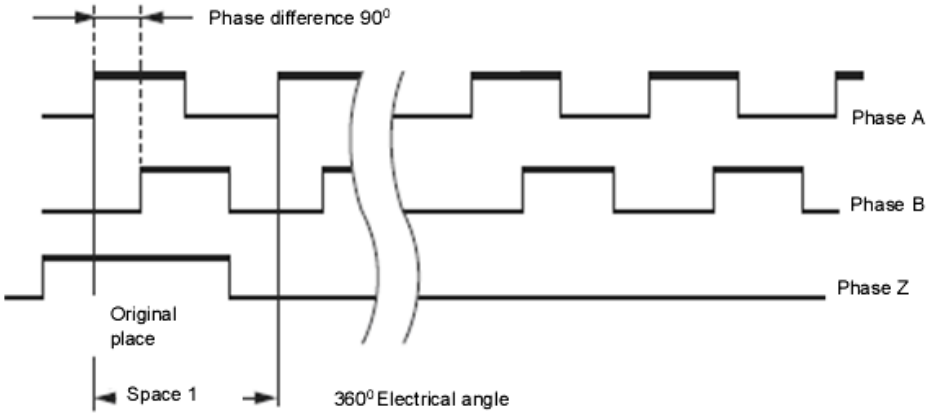
The application of the photoelectric encoder in the sorting unit



The structure of the incremental photoelectric encode



The diagram of the incremental encoder's output pulse



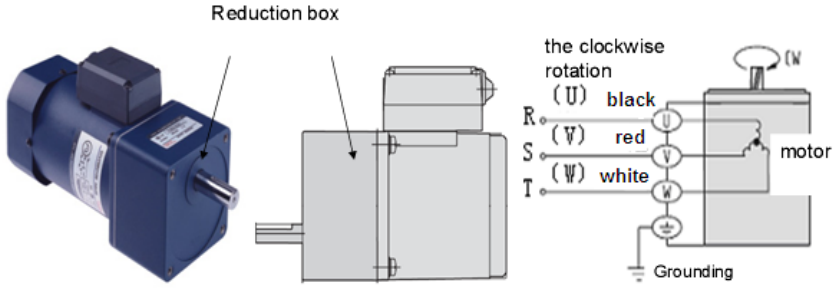
Task Two Control of Asynchronous Motor in the APL

Task objectives

1. Have a good command of the way of the asynchronous motor control；
2. To control asynchronous motor with the frequency converter；
3. Being capable of setting the parameter of frequency converter.

Subtask One Use of the AC Asynchronous Motor

The three-phase AC motor with reduction gearbox is used in the conveyor belt in XK-335B sorting station, so that the conveyor belt runs at the moderate speed

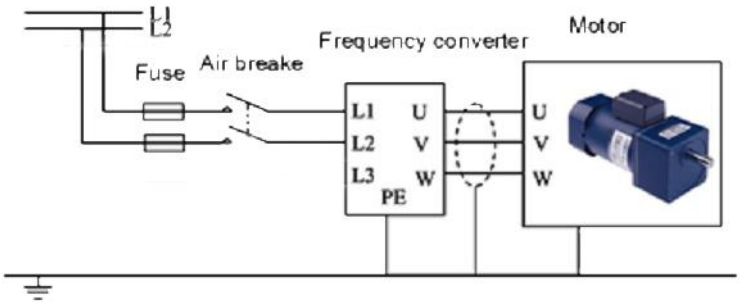


Three phase AC geared motor

It should be noted that during operation if one phase of the three-phase asynchronous motor is powered off, then it becomes single-phase operation. At this time the motor will still run in the original direction. However, if the load does not change and the three-phase power becomes the single-phase power, then the current will become bigger, causing the motor overheat. Special attention should be paid to this. If one phase of the three-phase asynchronous motor is powered off before starting, it cannot be started and we can just hear the humming sound. If it cannot be started after a long time, it will be overheated and trouble shooting is required immediately. Please note that the shell's grounding wire must be connected to the ground to prevent personal injury caused by electric leakage

Subtask Two Use of Universal Frequency Converter Driver

The speed and direction control of the three-phase AC geared motor in the sorting station in XK-335B adopts Powtrans' universal frequency converter PI9100. Its electrical wiring is shown in Fig. The three-phase AC power is output to the AC motor through the fuse, the AC contactor and the filter (optional).



Task Three Application of Servomotor & Driver in the APL

Task Objectives

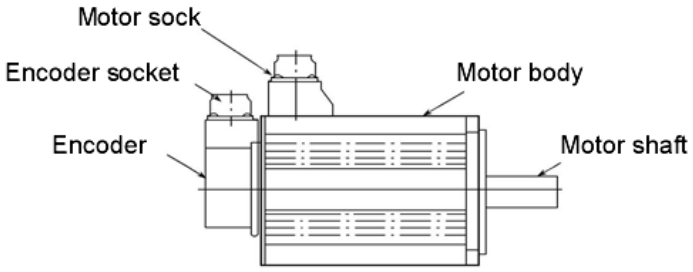
1. Grasp the characteristics and ways of servomotor control and the principles & electrical wiring of servo driver；
2. Being able to control a servomotor with a servo driver;
3. Being able to set the parameters of a servo driver.

Subtask One Getting to Know AC Servo Motor and Driver

In the delivery (transfer) unit of XK-335B, the Panasonic MHMD022P1U permanent-magnet synchronous AC servo motor is being used. The MADDT1207003 all-digital AC permanent magnet synchronous servo drive works as a movement control device of the transport manipulator, shown in the following picture.

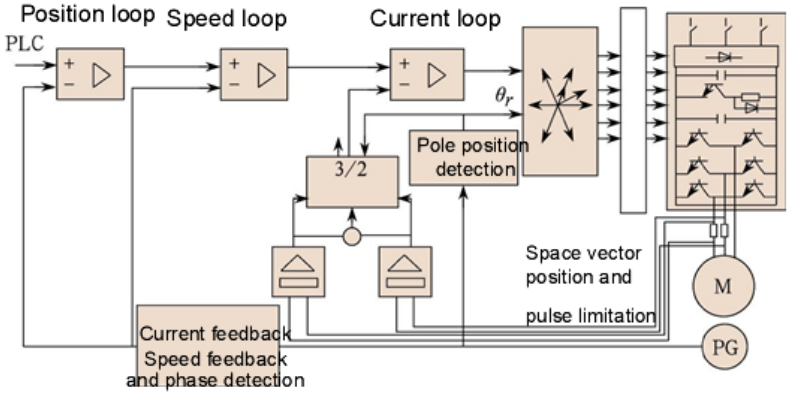


MHMD022P1U refers to: MHMD represents that the motor type is high inertia；02 represents that the rated power is 200 W; 2 means that the voltage specifications is 200 V; P means that the encoder is an incremental encoder, with the pulse number of 2500 p/r, resolution of 10,000, output signal lines of 5 .



The structure diagram servo motor

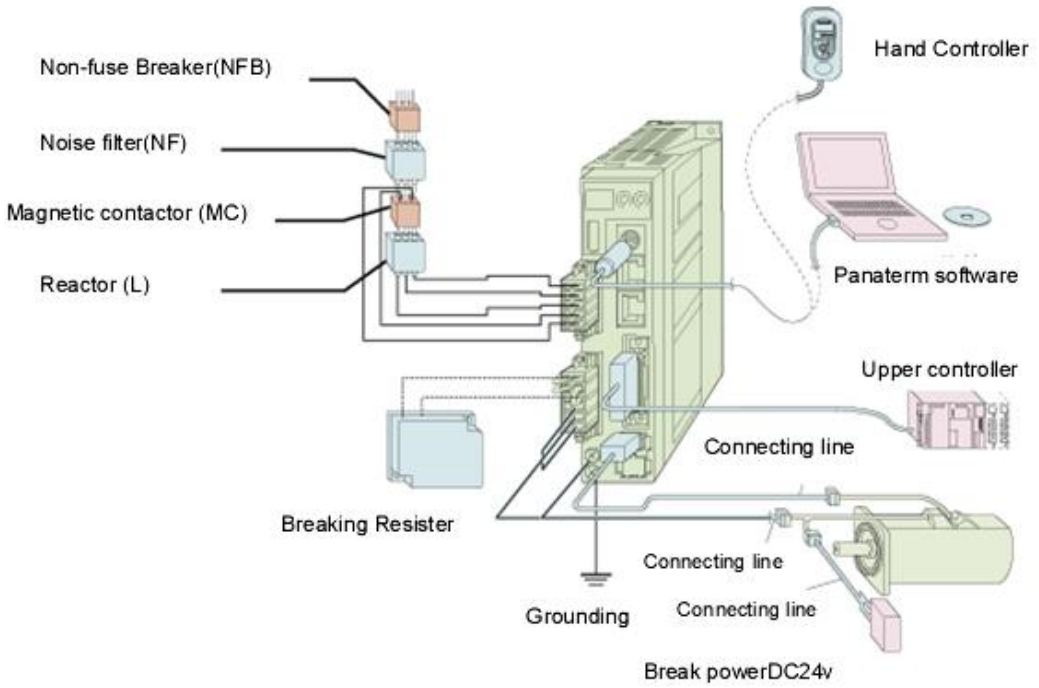
AC permanent magnet synchronous servo driver is composed of servo control unit, power drive unit, communication interface unit, servo motor and the corresponding feedback detection components. Its control system structure is shown in Fig. The servo control unit includes the position controller, speed controller, torque and current controller, etc.



Structure Diagram of the Servo Driver

Subtask Two Hardware Wiring of Servo Motor and Driver

The wiring between servo motors & drivers and peripheral devices are shown as the following picture. Input power flows through the circuit breaker, the filter, and then directly into the control power input (X1) L1C, L2C. The power following the filter flows through the contactor and the reactor into the servo driver's main power input (X1) L1, L3. Servo driver’s output power (X2) U, V, W connects with servo motor. The output signal from servo motor’s encoder should also be connected with the driver’s encoder input port (X6). The associated I/O control signal (X5) might also be connected with the PLC and other controllers as well. Servo drivers can also be connected with a computer or hand controller for parameter setting.



Wiring diagram between servo motor & the driver and the external devices

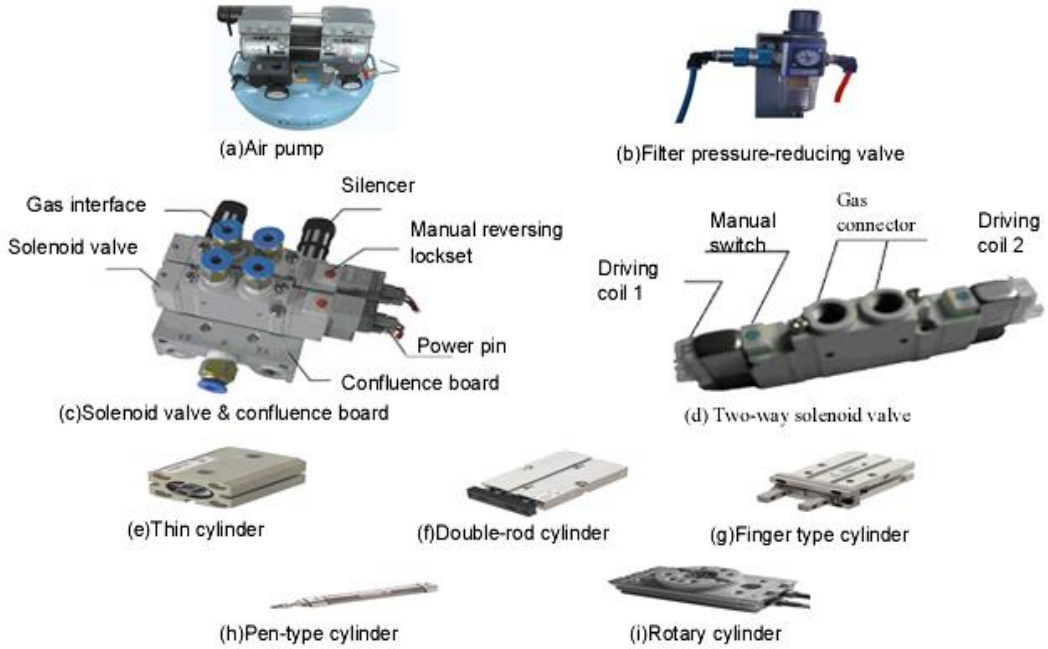
Task Four Application of Pneumatic Technology in the APL

Task Objectives

1. Master the functions and characteristics of the commonly used pneumatic components;

2. Being able to use the pneumatic components to form the pneumatic system and connect the gas path.

The pneumatic components in XK-335B



Subtask One Getting to know Pneumatic Pump

Air pump includes air compressor, pressure switch, overload safety protector, gas tank, pressure gauge, air supply switch, and main pipe filter.



Introduction to Air Pump Components

Subtask Two Getting to know Pneumatic Actuating Components

The commonly used actuator components in pneumatic system are cylinders and gas motors. Cylinder is for the linear reciprocating movement; gas motor is used for continuous rotary movement. In XK-335B, only cylinders are used, including the pen type cylinder, thin cylinder, rotary cylinder, double-rod cylinder, and finger cylinder.

Cylinders used in XK-335B



(a) Thin cylinder



(b) Double-rod cylinder



(c) Finger cylinder



(d) Pen-type cylinder



(e) Rotary cylinder

Classification

Mini cylinder: 2.5-6mm

Small sized cylinder 8-25mm

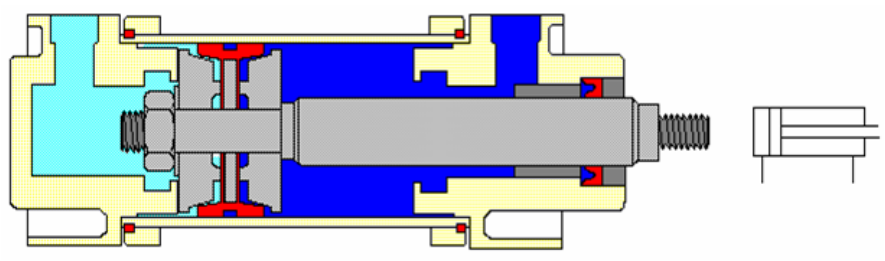
Middle sized cylinder: 32-320mm

Way of installation: Fixed cylinder and swing cylinder

Way of Iubrication: Oil feeding cylinders and non-oil feeding cylinders

Driving mode: Single-acting cylinders and double-acting cylinders

A cylinder mainly consists of cylinder body, piston rod, front and back lids, seals and so on



Structure of a common single-piston dual-function cylinder

Pneumatic Swaying Platform



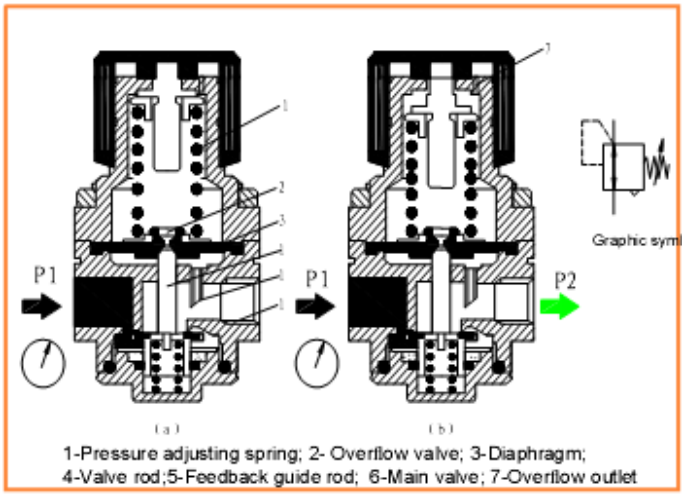
Subtask Three Getting to know Pneumatic Control Components

The pneumatic control components used in XK-335B involves pressure control valves, direction control valves, flow rate control valves according to the use and function.

1. Pressure control valve

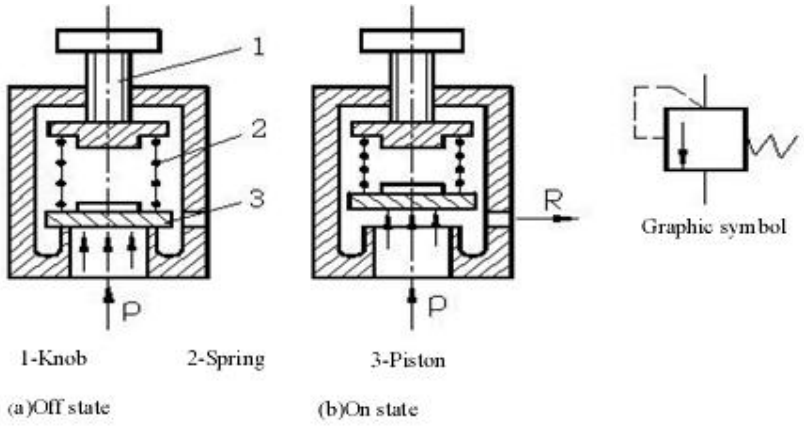
The pressure control valves used in XK-335B mainly include reducing valves and overflow valves.

（1）Pressure-reducing valve : It is used to reduce the pressure from the air compressor to meet the need of each pneumatic device and to keep the pressure stable. The following figure is a direct pressure-reducing valve.





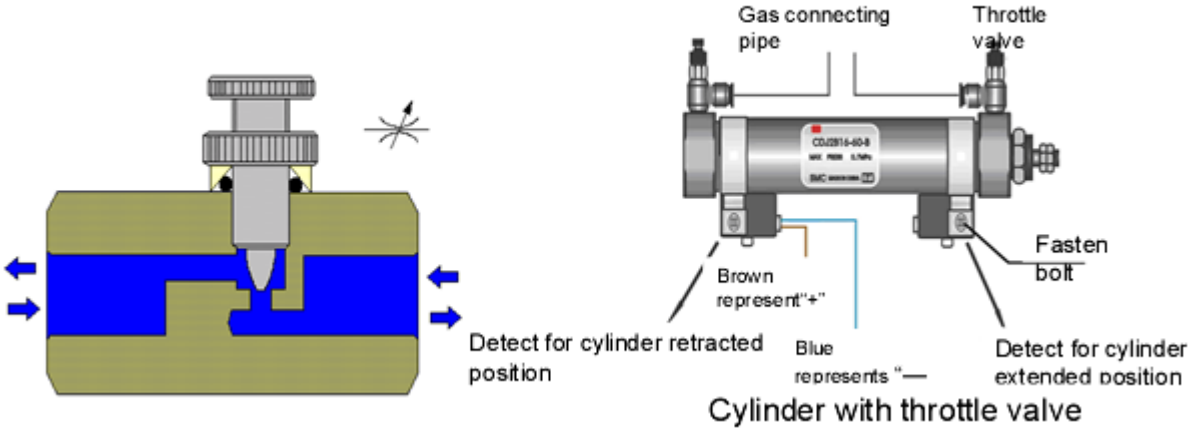
（2）Overflow valve The function of the overflow valve is to exhaust gas automatically in order to reduce the system pressure so as to ensure the system safety when the system pressure exceeds the set value. For this reason it is also called the safety valve



Functional Diagram of Safety Valve

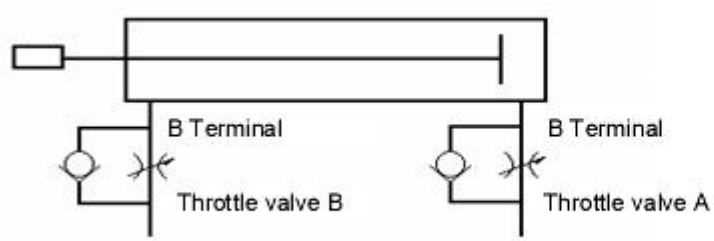
2. Flow Control Valve

Throttle valve reduces the air flow cross-section so as to increase the air flow resistance, and accordingly the air pressure and flow would decrease. There is an adjustment screw on the valve body, you can adjust the open of the throttle valve (stepless regulation), and maintain the same open. This kind of valve is called adjustable open throttle valve

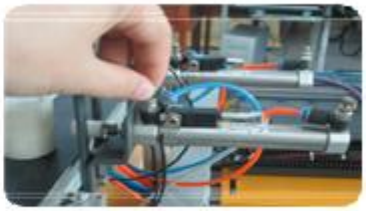


Cylinder with throttle valve

The following figure is a diagram of connection & adjustment of a double action cylinder with two limited stretch throttle. When you adjust the throttle valve B, the extending speed of the cylinder will be adjusted. When you adjust the throttle valve A, the retracting speed of the cylinder will be adjusted.



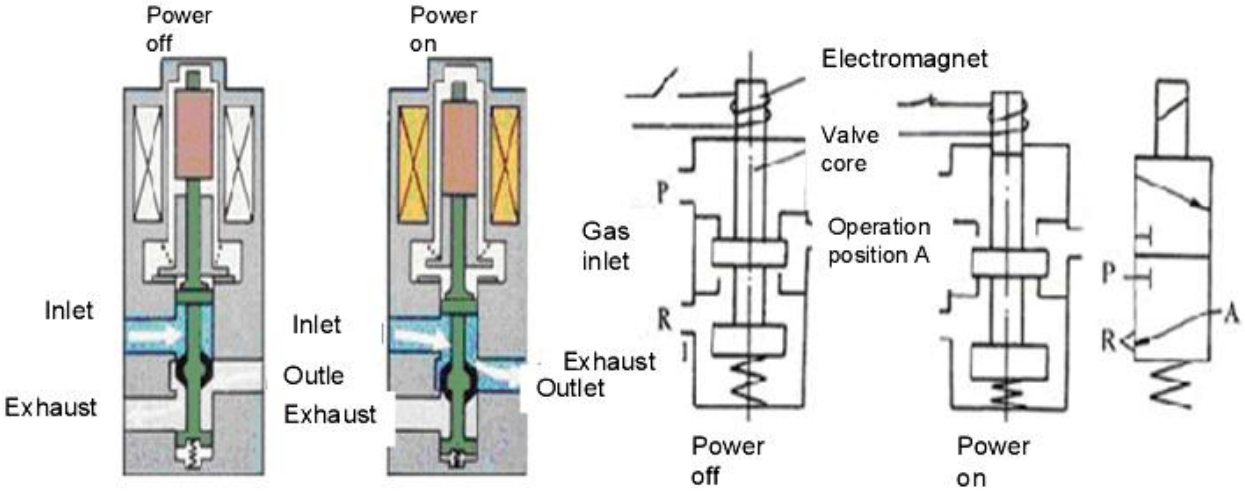
(a) Throttle valve connecting and adjusting diagram


(b) Actual adjustment

3. Direction control valve

It is used to change the air flow direction or to determine the on-off state. Usually solenoid valve is used.

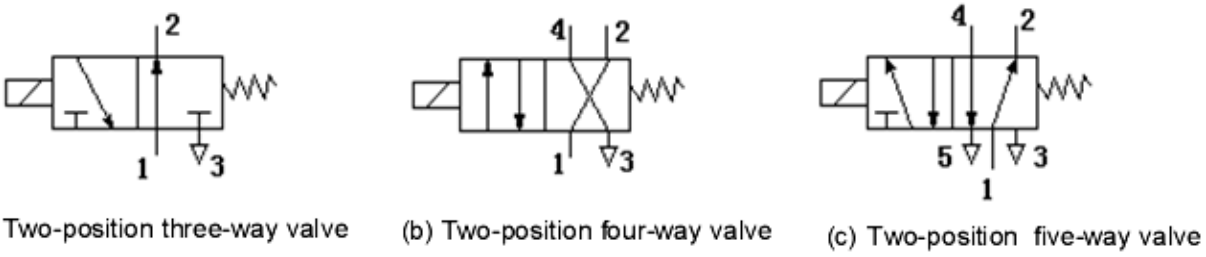
Solenoid valve is used to change the air flow direction by way of when the electromagnetic coil being charged with electricity, static magnetic iron producing electromagnetic suction to the dynamic iron core, resulting in the valve core switching.



Working principle of single electrical control reversing solenoid valve

Concept of "position"

The so-called "" refers to change the air direction and valve core different operation positions with relative way to the valve body. "way" refers to the passes between the direction control valve and the system. The number of the port determines the "how many ports". There are only two operation positions and also there is gas inlet P, the operation port A and exhaust port R, so it is called two-position three-way valve. The following picture lists the graphic symbols of two-position three-way, two-position four-way and two-position five-way single electrical control reversing solenoid valve. The number of square in the diagram stands for the number of position, and the "┬" and "┴" symbols in the square indicate that the interfaces are mutually independent.



(a) Two-position three-way valve (b) Two-position four-way valve (c) Two-position five-way valve

Graphic symbols of some single control reversing solenoid Valves

Task Five Application of PLC in the APL

Task Objectives

1．Grasp PLC working principle, features of external interface, choosing principle of input and output, common commands;

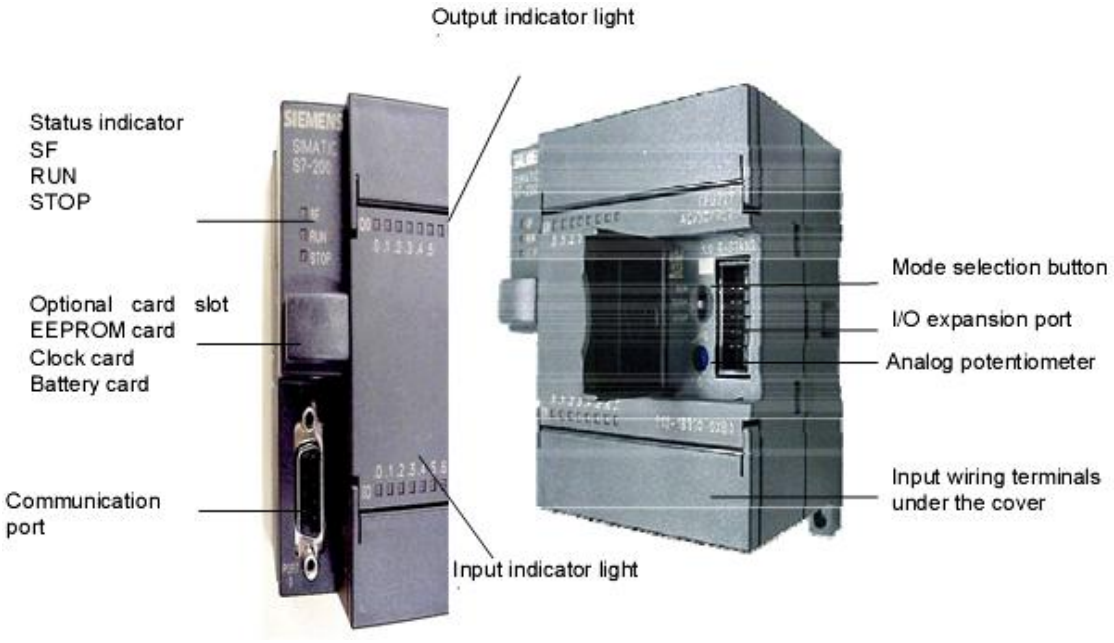
2．Analyze the process requirements of the control system, determine the number of input & output of digital and analog

3．Use common commands and to make control system program

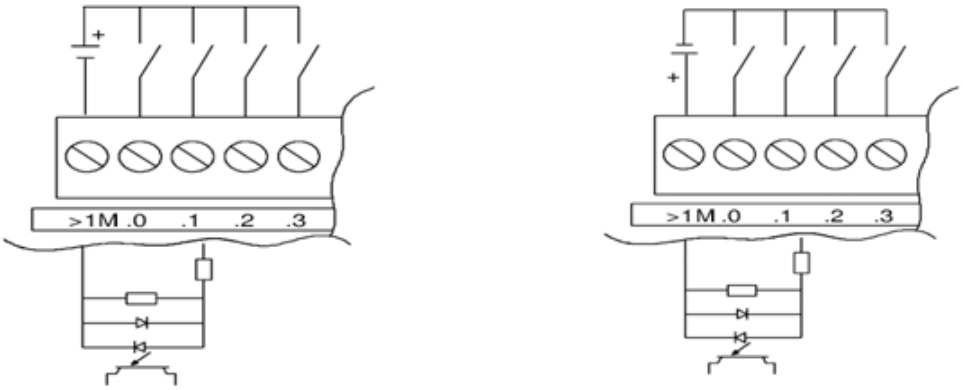
Subtask Getting to know the structure of Subtask Getting to know the structure of S7-200 PLC S7-200 PLC

S7-200 series PLC belongs to hybrid PLC, which is composed of PLC host and expansion modules. The PLC host is composed of some basic modules like CPU, memory, communication circuit, basic input & output circuit and power. It is an integral PLC and can realize the control function individually. It includes the minimum composing unit needed by a control system. Following Fig. is the shape structure of S7-200 CPU module, which combines a microprocessor, an integrated power and digital I/O in a compact enclosure.

Shape structure of S7-200CPU module



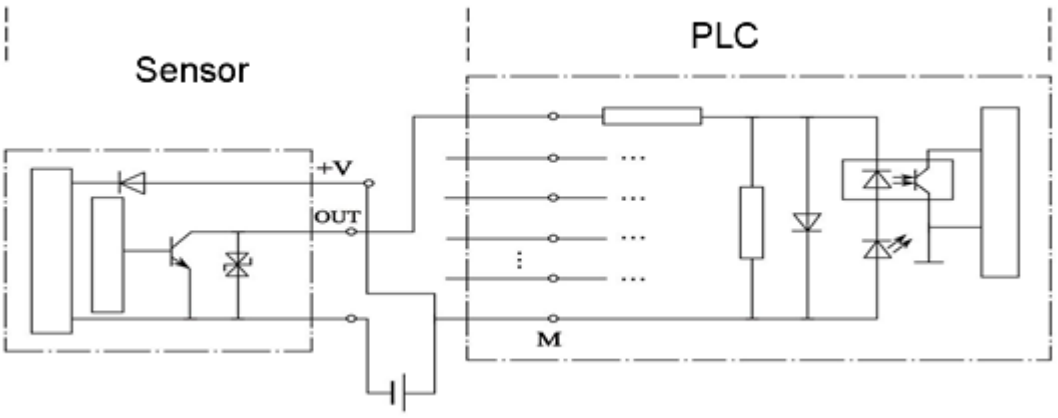
Switching Input/output interface



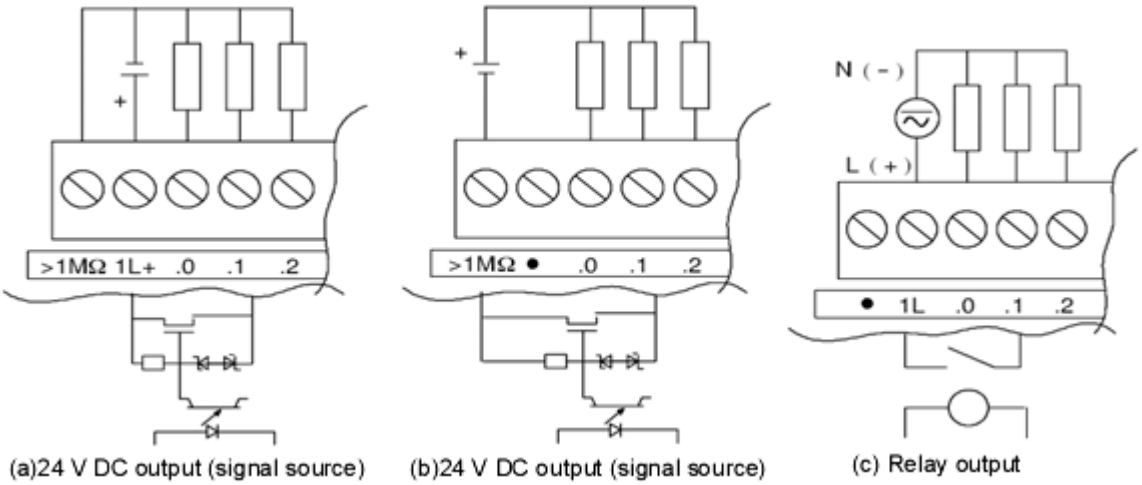
(a)24VDC input as sink input

(b)24VDC input as source input

Connection between NPN output sensor and PLC



Wiring diagram of S7200PLC output mode



Task Six Application of Communication in the APL

Task Objectives:

1. Grasp PPI communication interface protocol used in PLC and network programming instructions;

2. Grasp the installation, programming and testing of PPI communication network

In modern APL, the controlling devices in different working stations do not work independently. For example, the five working stations work as a whole in XK-335B by way of communication means and information interexchange to form an integration and finally to improve the control and reliability of the equipment to realize “Central Handling and Separate Control”.

Subtask Getting to know PPI Communication

What is PPI Communication?

1. Basic knowledge of communication?

The function of communication technology is to realize data interchange between different devices. PPI (point to point) is the serial communication from point to point. Serial communication transmit a binary number once. Thus the transmitting speed is slow. Because the wiring is short, it can transmit data in long distance. PLC-200 has its own serial communication interfaces.

Task Seven Application of Human-Machine Interface & Configuration in the APL

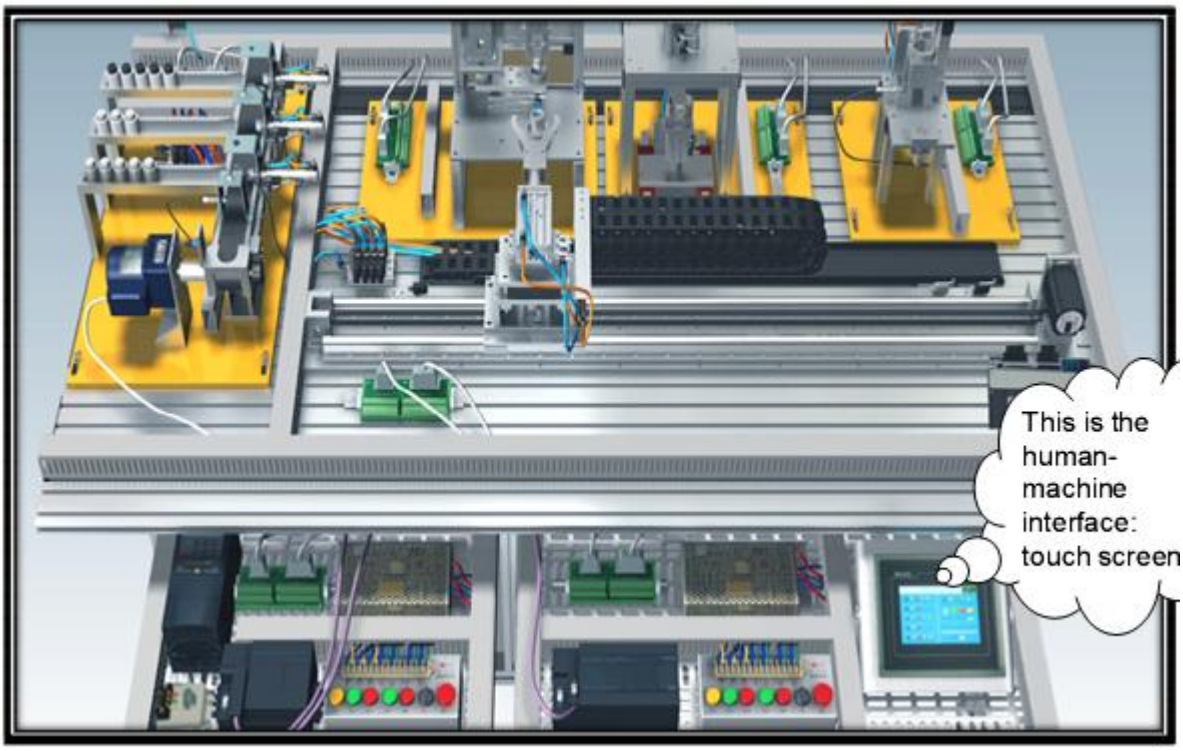
Task Objectives

1. Grasp the concept and characteristics of human-machine interface, and its configuration method;

2. Being able to write configuration program of the human-machine interactive configuration program, and to perform installation & testing.

Subtask One Getting to know TPC7062K human-machine interface and MCGS embedded industrial control configuration software

TPC7062K is a high-performance embedded integrated industrial control machine with low-power consumption and CPU centered. The design of the product uses a 7-inch high brightness TFT LCD screen (resolution ratio 800 × 480), four-line resistance touch screen (resolution ratio 4096 × 4096). Meanwhile it also pre-installed embedded real-time multi-task Microsoft operating system-WinCE.NET (Chinese version) and MCGS embedded configuration software (running version).MCGS embedded configuration software is developed by Kunluntongtai Company and adopted in mcgsTpc series human-machine interface devices. It can mainly complete local data collection, monitoring, frontier data processing and control.



XK-335B Automatic Production line

1. Simple use of TPC7062K



2. Getting to know MCGS embedded configuration software

Combining with other related hardware, MCGS embedded configuration software is capable of quickly and easily developing devices for field acquisition, data processing and control devices (equipment). It can configure a variety of special devices such as: intelligent instruments, data acquisition module, paperless recorder, non-attendant field gathering station and human-machine interface.

Main Functions of MCGS embedded configuration software

①Simple and flexible visual operating interface with all-Chinese and visual development interface meet Chinese habits and requirements.

②It is a real 32-bit system with real-time and better parallel processing properties and parallel processing tasks in the unit of line stroke.

③Plentiful and vivid media images: providing the operators relevant information in the way of images, icons, report forms, curves and so on.

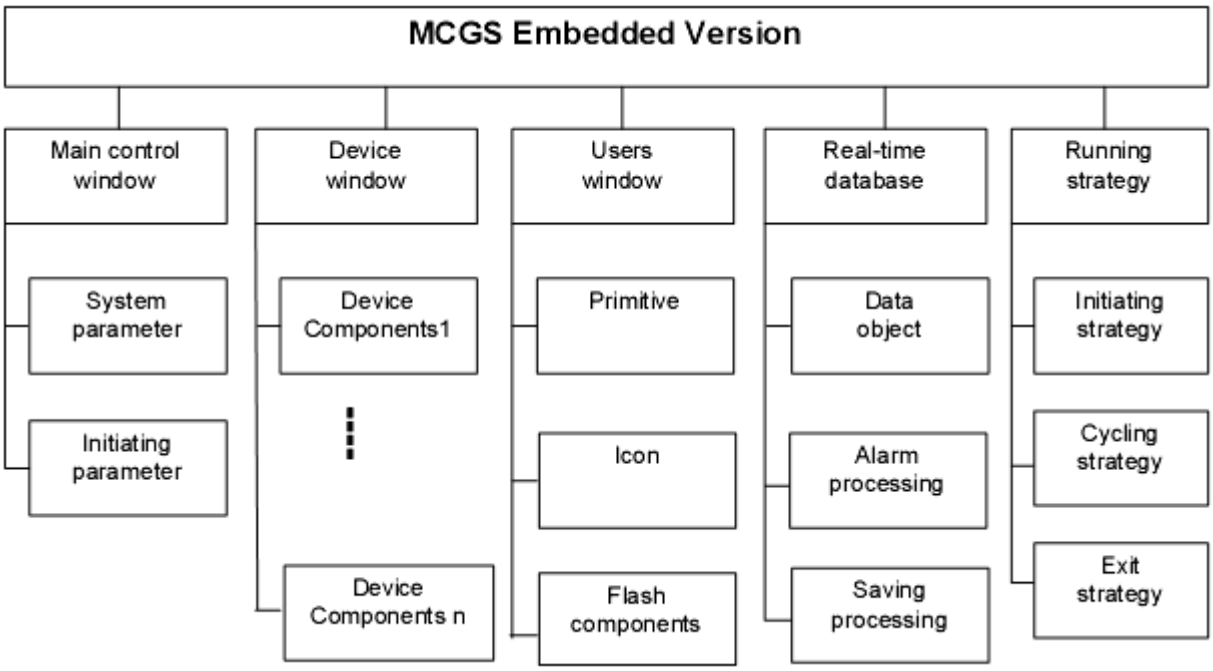
④Perfect security system: providing good security system and setting different operating limits of authority for users of different levels.

⑤Strong network function: powerful network communication function.

⑥Variety of alarm functions: providing different alarm modes, being with various alarm types, convenience of setting alarm for users.

⑦Supporting a variety of hardware devices.

In short, MCGS embedded configuration software has the same great functions as that of the general configuration software. It is easy to operate, learn and to use.



The composition map of MCGS embedded configuration software

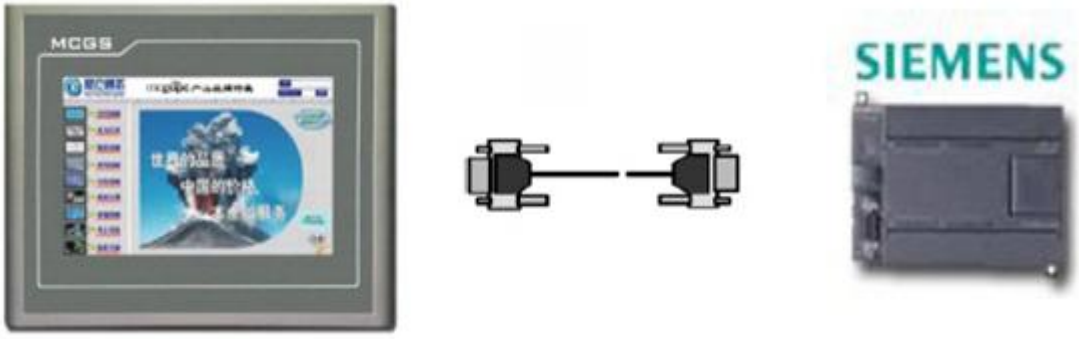
Connection of TPC7062K with configuration computer



Subtask Two Wiring of TPC7062K & PLC and Engineering Configuration

1. Wiring of TPC7062K with PLC

After getting to know about TPC7062K, we come to the communication ways of it with Siemens S7-200PLC. The way of wiring is shown in the following Fig.



2. Configuration of connecting MCGS embedded version with Siemens S7-200P

(1)Project Establishment

(2) Engineering Configuration

Chapter Three Projects Acceptance

Task Objectives

1. Being able to conduct installation and testing for each station in the Automatic Production Line;

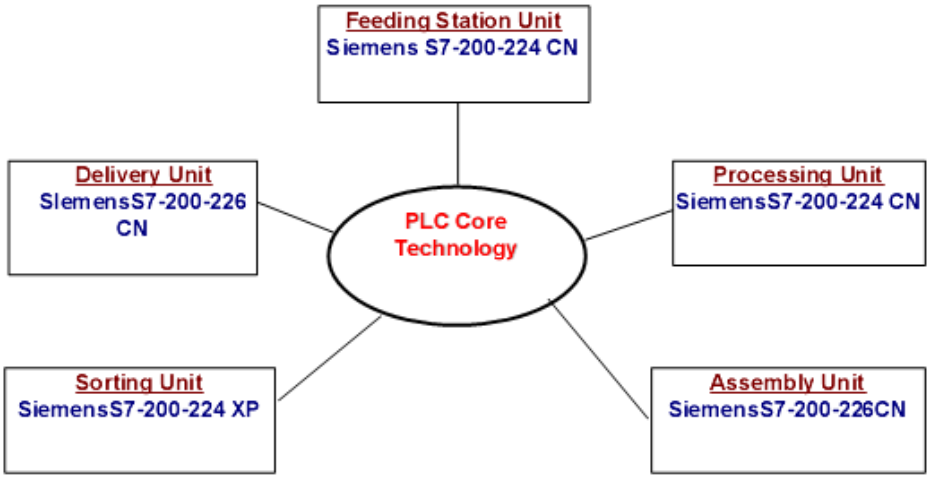
2. Being able to do the control programming and testing for each station in accordance with the control requirements;

3. Being able to solve problems during installation and operation of the Automatic Production Line.

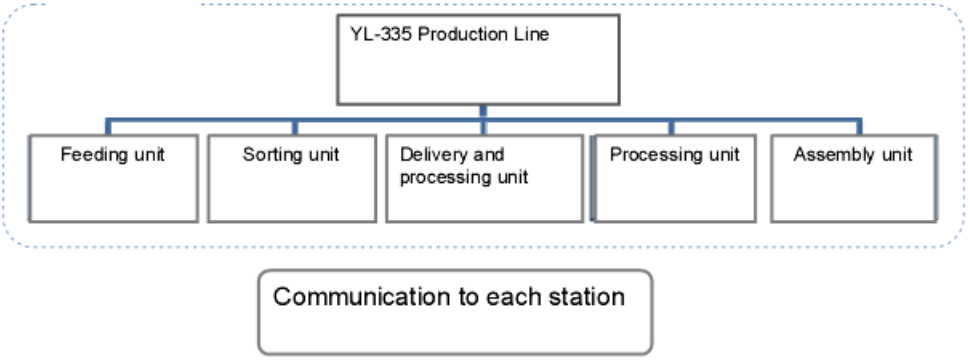
Training Mode

XK-335B Automatic Production Line comprises 5 stations. Each of them can run as an independent unit or can be combined with any others, which represents the application of PLC in different working environment, different applied field, different applied aging situation. Each station has a main technical unit, together with some other technical units. Reuse of PLC technology in different situations indicates its key position in the electromechanical control field. It represents course development concept of integrated teaching environment and PLC core technology.

Integration of PLC Core Technology with Teaching Environment



Teaching Mode



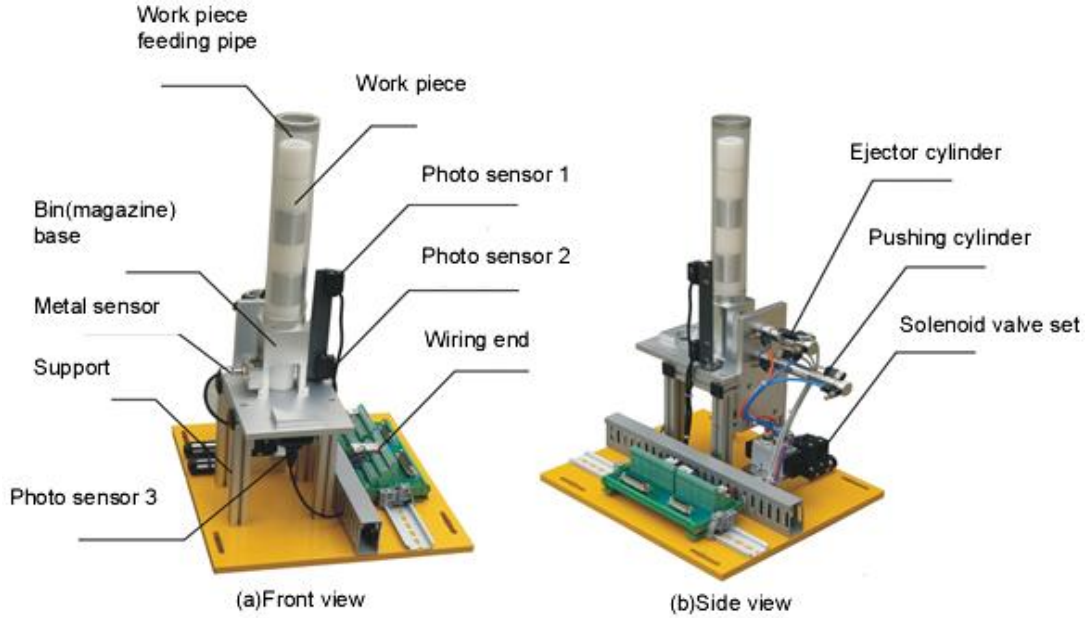
Tasks

1. Installation & Testing of the Feeding Unit
2. Installation & Testing of the Processing Unit
3. Installation & Testing of the Assembly Unit
4. Installation & Testing of the Delivery Unit
5. Installation & Testing of the Sorting Unit

Task One Installation and Testing of the Feeding Unit

Getting to Know the Feeding Unit

The feeding unit is the starting unit in the Automatic Production Line, which supplies materials to other units in the system. It is equivalent to an automatic feeding system for a real production line. Its main structures are feeding pipes for work pieces, work piece pusher device, support, valves set, terminal board assembly, PLC, emergency stop button and start/stop button, cable trough, base plate, etc.



Overview of the Feeding Unit

Functions of the Feeding Unit:

To push work pieces (raw material to be processed) in the feeding bin (magazine) automatically out onto a charging table as needed, so that a manipulator on the delivery unit will pick it and deliver it to other units.

Operation Process of the Feeding Unit

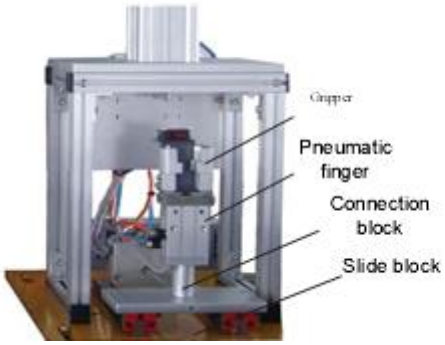
Work pieces are vertically piled in the feeding bin (magazine). The pushing cylinder is located at the bottom of the feeding bin (magazine), and its piston rod can move through the bottom of the feeding bin. When the piston rod is back at home position, it is at the same horizontal level as the lowest work piece, and the clamping cylinder is at the same horizontal level as the sub-lowest work piece. When work pieces are needed to be pushed onto the charging table, the piston rod of the clamping cylinder

is first to be pushed out and to hold sub-lowest work piece tightly. Then the piston rod of the pushing cylinder is pushed out and to push the lowest work piece onto the charging table. After the pushing cylinder gets back and pulls out from the bottom of the feeding bin, the clamping cylinder can then returns and releases the sub-lowest work piece. In this way, the work pieces in the feeding bin will move downward by gravity another one ready to be pushed out.

Task Two Installation and Testing of the Processing Unit

Getting to Know the Processing Unit

The function of the processing unit is to take work pieces to be processed from the material table to a place right below the stamping cylinder for stamping in the processing area, and then bring the work pieces back to the material table.





Operation Process of the Processing Unit

The material table is used to fasten a work piece to be processed and take the work piece to the place right below the processing (stamping) unit for stamping. It mainly consists of clamp, pneumatic fingers, extension & retracting cylinder, linear guiderail and its sliding block, magnetic inductive proximity switch and diffusion photoelectric sensor. Following normal working of the system, initial status of the sliding material table is that the stretching & withdrawal cylinder extends and the pneumatic finger of the material table opens. The processing procedure is: as the delivery (transferring) unit has delivered the material onto the material table and the sensor has detected it, PLC control program then instructs the pneumatic finger to clamp the work piece→the material table moving back to the place right below the stamping cylinder in the processing area→piston rod of the stamping cylinder stretching down for stamping the work piece→retract upward after the action of stamping→material table reaches out again→when in place, the pneumatic fingers loosen, and the processing is complete.

Then the signal of completion is sent to the system; the preparation is made for the next processing.

Task Three Installation and Testing of the Assembly Unit

Getting to Know the Assembly Unit

The assembly unit is designed to assemble two different sizes, small cylindrical work pieces together, i.e. taking small cylindrical work pieces out of the feeding bin into a center hole on the material table. The Assembly unit mainly consists of simple feeding bins (magazine), feeding devices, swing material tables, manipulator, locating devices for semi-products, a pneumatic system and its valve sets, a signal collector and its auto-control system, components of terminals board for electric connection, and in addition, status signal indication lights for the whole production line, aluminum supports and bases for installing other devices, supports for installing sensors and other auxiliary devices.

Task Four Installation and Testing of the Sorting Unit

Getting to know the sorting unit

Sorting unit is the last unit in the Automatic Production Line, in which the processed and assembled work pieces sent from the last unit should be sorted and work pieces of different colors from different feeding chute should be distributed. When the photoelectric sensor on the feeding end detects work pieces on the conveyor belt which are delivered from the transport station, the frequency converter is started and work pieces will be delivered into the sorting area to be sorted.

Sorting unit mainly consists of the delivery (transport) and sorting devices, drive device, frequency converter module, the solenoid valve set, the wiring terminal, PLC module, the base plate and so on. Delivery (transport) and sorting devices are supposed to transport the processed and assembled work pieces and sort them after being detected by the metal sensor and the fiber optic sensor. It is mainly composed of the conveyor belt, feeding chute, material pushing cylinder, the diffuse-type photoelectric sensors, rotary encoders, metal sensors, fiber optic sensors and magnetic proximity sensors.

Task Five Installation and Testing of the Delivery Unit

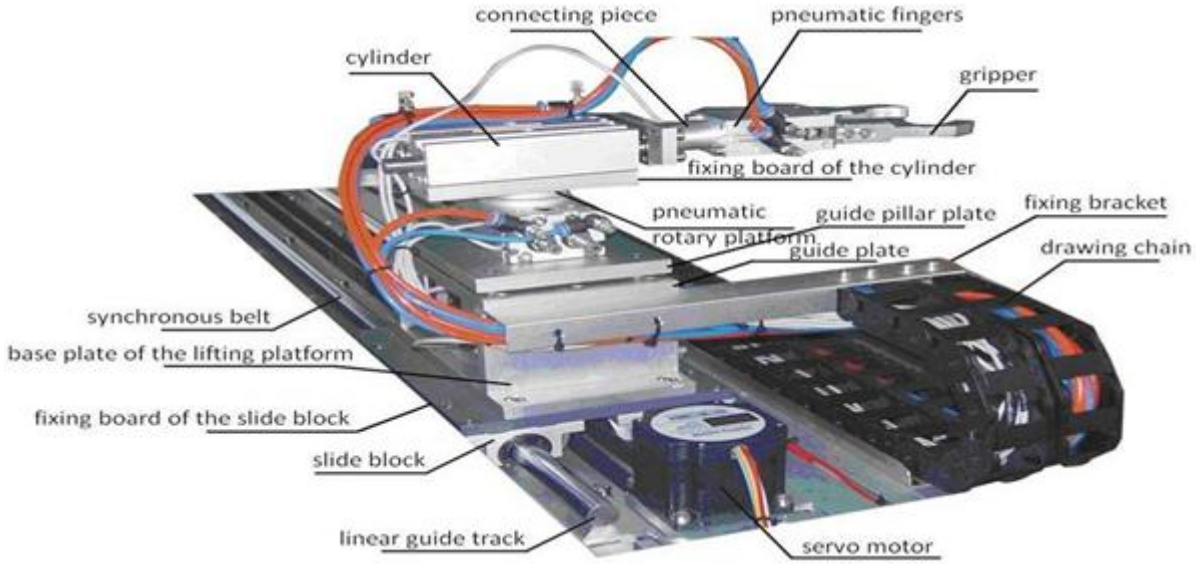
Getting to Know Delivery Unit

The delivery unit is the most important and the most heavy-loaded part of the Automatic Production Line. In this unit, the gripping manipulator is driven to the precisely position of the material table and grip work pieces from the material table to transport it to the specified place and drops it.

Gripping manipulator device

The gripping manipulator is a working unit that can realize four degree of freedom (the four-dimensional movements of lifting, stretching & withdrawing, pneumatic finger clamping/releasing and moving along the vertical shaft) movements. The device is installed on the slide board of the

servo drive components. Driven by the drive components, it reciprocates in straight line and moves to the material table of other working units and finishes the function of gripping and releasing work pieces.



Specific composition is shown as follows:

①The pneumatic finger: A double acting cylinder is controlled by a two-position five-way bidirectional electrical control valve. It is used to grip & delivery the work piece in every working station. The working principle of bidirectional electrical control valve is like the bi-stable trigger. That is, the output state is determined by the input state. Once the output state is confirmed, even there is no input, the bidirectional electrical control valve still maintains the pre-triggered state.

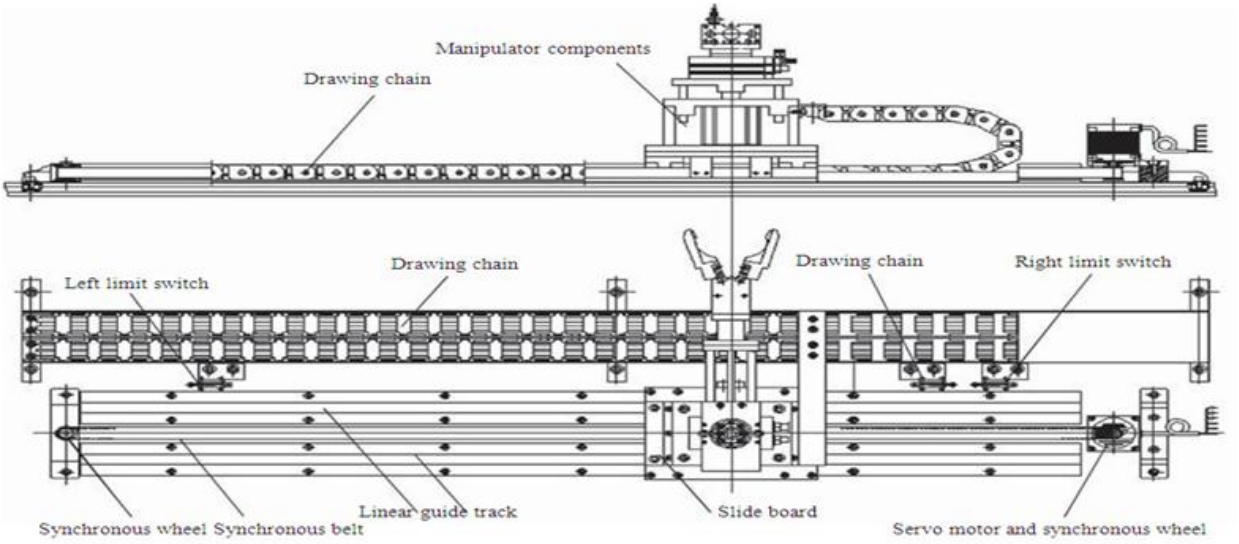
②Double-rod cylinder: The double acting cylinder is controlled by a two-position five-way single control valve, which is used to control the gripper's stretching & withdrawing.

③Rotary cylinder: Double acting cylinder is controlled by a two-position five-way single control valve, which is used to control the arm's rotation of 90 degrees forward or backward. The cylinder's rotation ranges from 0 to 1800 and the adjustment is realized by two fixing buffers below the throttle valve.

④Lifting cylinder: Double acting cylinder is controlled by a two-position five-way single control valve, which is used to control the manipulator's lifting. The running speed of cylinders above is controlled by the throttle valve's adjustment of intake flow rate in the air inlet.

Servo drive components

Servo drive components are used to drag gripping manipulator for reciprocating linear movement to fulfill the precise positioning. The following figure shows the front and top view of the components, in which the gripping manipulator has been installed on the slide board of the components.



The drive components are composed of the servo motor, synchronous wheel, synchronous belt, linear guide track, slide board, drawing chain, the origin switch, the left limit switch and the right limit switch.

Driven by the servo driver, the servo motor brings along the slide board through the synchronous belt and the synchronous wheel to reciprocate in the linear direction along the linear guide track, and thus brings along the gripping manipulator fixed on the slide board for reciprocating linear movement.

All the air pipes and wires on the gripping manipulator are placed along the drawing chain and connected to the solenoid valve set and the connecting terminal block after entering the line groove.



Chapter Four: Project Decision

Task Objectives

1．Being able to finish the Installation of Automatic Production Line within the set time；

2．Being able to implement touch screen interface set-up, web assembly and control program design of each single station according to the work commitments;

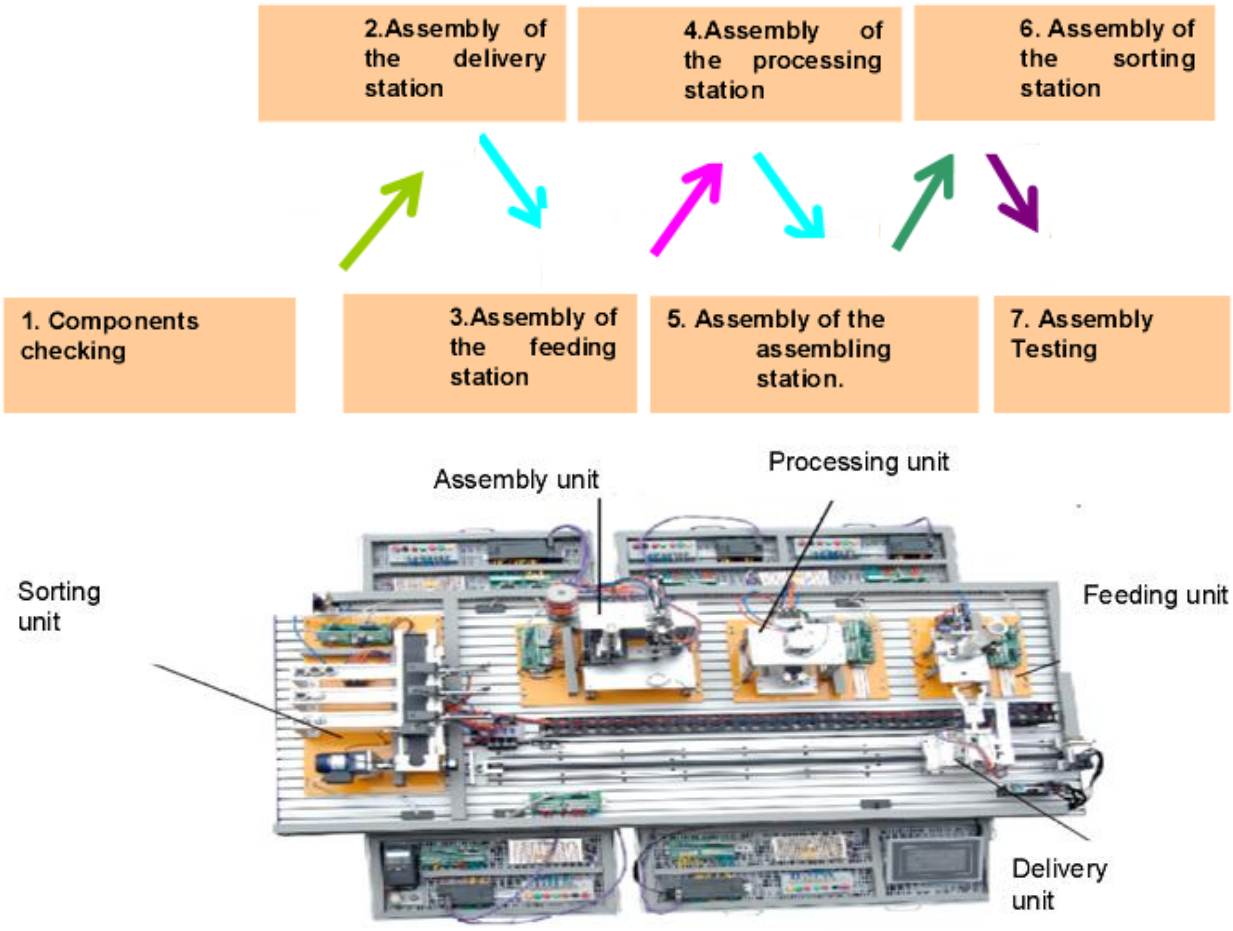
3．Being able to resolve the common problems during the installation and operation of Automatic Production Line；

4．Being able to obtain Programmable Control System Designer Professional Qualification Certificate (Level 3)

Task One Equipment Installation of XK-335B APL

(1)First of all, you are supposed to check whether there are all necessary components according to manual. And then check if the quality and condition of the components meet the requirements.
(2)Then you should finish part of the assembly work for feeding, processing, assembling, sorting and delivery units.
(3)Install them on the work table of XK-335B.

Installation step:



Work table of APL after the installation

Task Two Gas Circuit Connection in XK-335B APL

Complete the gas circuit connection of XK-335B APL based on the gas circuit knowledge learned in Chapter 2, related training of gas circuit connection of each station in Chapter 3 and the control requirements of work commitments.

Task Three Electric Circuit Design and Connection in APL

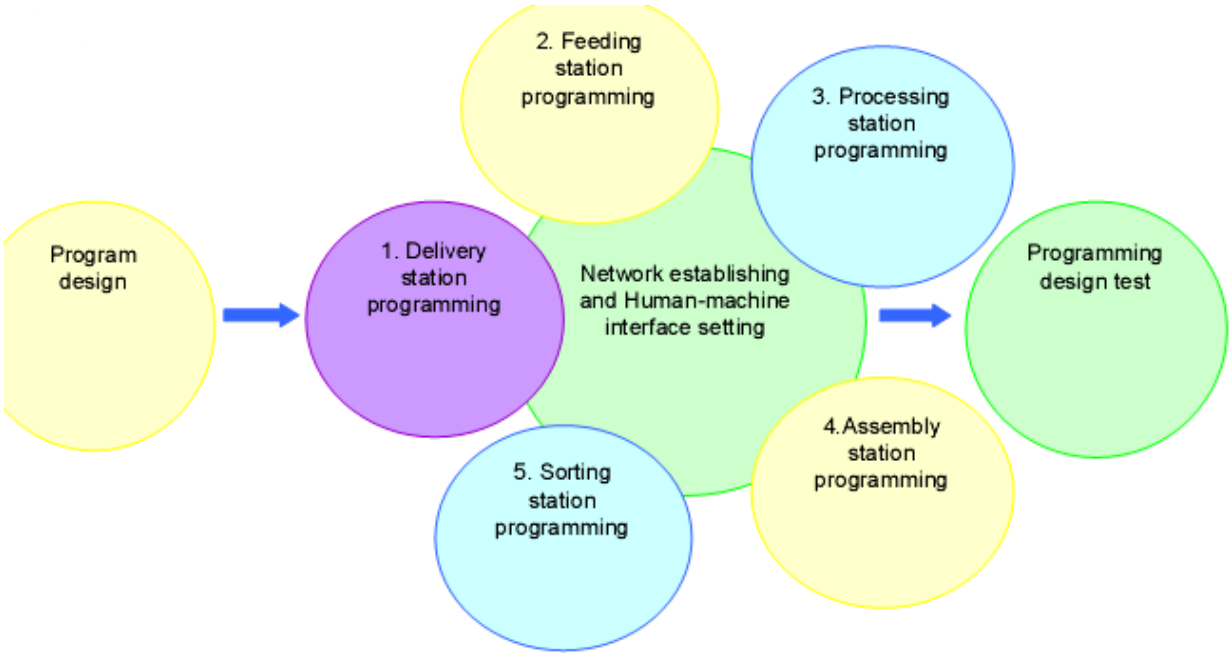
According to the control requirements in work commitments, you are supposed to design the control circuit of the APL and connect electrical devices according to the specified PLC I /O address.

Hardware configuration capacity requirements of the system to obtain PCSDPQC

Task	Competence	Requirements
Equipment Selection	1. Being able to select the PLC model, according to the capacity of input / output point, program capacity and surface sweep speed; 2. Being able to select the switching input/output unit according to the technical specification; 3. Being able to select the analog input/output unit based on technical specification and set the hardware; 4. Being able to select the suitable peripheral equipments of switching and analog unit and set the hardware 5. Being able to calculate system power according to its configuration and select the PLC power unit and external power.	1.Selection principle of PLC model 2.Selection principle of switching input/output unit 3.Selection principle of analog input/output unit 4.Selection principle of PLC power unit.
Reading& equipment Installation of hardware diagram	1. Being able to read electric diagram; 2. Being able to read wiring diagram; 3. Being able to read components layout; 4. Being able to read components' site map; 5. Being able to carry out the site installation of digital quantity and analog quantity in accordance with the requirements of the drawing.	1.Electric graphic symbols and drawing specifications 2.Technical requirements of electric wiring. 3.Basic knowledge of installation & construction of electric equipment.

Task Four Programming and Program Test

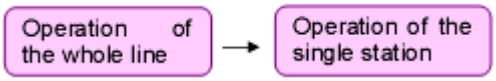
Now you are supposed to establish the network following the requirements given in work commitments to achieve the data transmission between five programmable controllers. To realize each control task required in commitments based on the five stations program design. Master certain programming development ability.



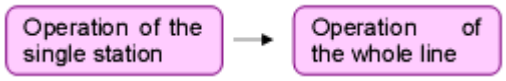
Designing ability of the system to obtain PCSDPQC

Task	Competence	Requirements
Project analysis	1. Can analyze the process requirement of the controlled object in control system composed of digital quantity and analog quantity; 2. Can determine the parameters of the switching and analog quantity in single control system composed of digital quantity and analog quantity. 3. Can conduct statistics of switching input/output points and analog input/output points of single device control system composed of digital quantity and analog quantity. And conclude the technical specification.	1. Types of controlled object; 2.Basic knowledge of the switching; 3. Basic knowledge of analog quantity
Design of control plan	1. Can design block diagram of single device control system composed of digital quantity and analog quantity. 2. Can design the flow diagram of single device control system composed of digital quantity and analog quantity.	1. Basic principle and requirements of the design of PLC control system. 2. Graphic symbols and drawing rules for flow diagram of the design of PLC control system.

Task Five Testing and Fault Analysis in the APL



1. After completing the work cycle, the selection switch on the human-machine interface changes to the single station mode.
2. The work mode of button/indicator module in each station selectors is on the single station mode.
3. When a work unit becomes an independent system, the manual control mode is on. The command signals for equipment operation and the status signal for running are both given and showed by button/indicator module in the work unit.



1. Each work station is off. Work mode of button/indicator module in each station is on the whole line operation mode.
2. The mode on the human-machine interface switches to the whole line operation mode, the system enters the whole line operation mode.
3. The main command signals (reset start, stop and etc) of the APL system running are given through the touch screen human-machine interface

Meanwhile, human-machine interface also displays system running status information.

Chapter Five Project Challenging

Task Objectives

1. Understand the function, usage and features of PROFIBUS.
2. Understand the function, usage and features of IPC configuration software.
3. Understand the function and features of the Industrial controller.
4. Understand the function, usage and features of robot.

Task One PROFIBUS Technology

Subtask One Getting to Know PROFIBUS

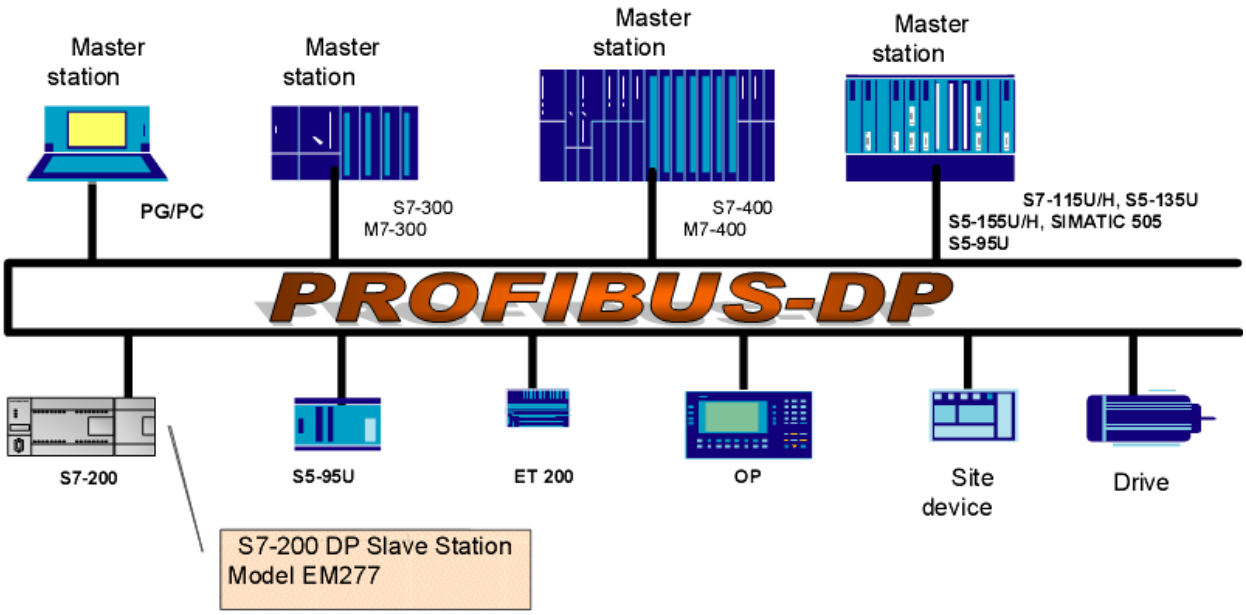
—PROFIBUS is the leading and open field bus system

—Independent market survey shows that PROFIBUS is the leading technology in European market, with the highest increasing rate in the world. There are over 1,500,000 basic PROFIBUS equipment installed in 1997, with annual growth about 20% to 30%.

Application Field of PROFIBUS



Siemens PROFIBUS Communication Network



Subtask Two Getting to Know the Basic Functions of PROFIBUS

PROFIBUS is composed of three compatible parts:

PROFIBUS-DP: It is mainly used for the communications between master station and slaver. That is the communication between the CPU and remote stations.

PROFIBUS-FMS: It is mainly used between master station and master station. That is the

communication between the CPU and PC. It can also communicate with other producer's equipment.

PROFIBUS-PA: It is laid in the production site. Serial connection is located on the forefront devices (equipment) and apparatuses for production process monitoring.

MAIN FUNCTIONS OF PROFIBUS-DP

Transmission technology	RS－485 double strand, double cable or optical fiber cable. Baud rate is from 9.6kbit/s to bit/s.
Synchronization	The control command allows the synchronization of input and output. Synchronization mode: synchronized output; Lock mode: synchronized input
Operation mode	Operation-clear-stop
Bus access	Command transmission among each master station. Master-slave transmission is between master station and slave station. Support single master or multi-masters system. The biggest number of stations (master-slave devices) in the bus is 126
Communications	Point to point(client data transmission) or broadcast(control command).Circulation data trans mission of master-slaver client and non-circulation data transmission of master-slaver client
Reliability	All information transmission is carried out on Hamming distance HD=4.DP slave station has t he Watchdog Timer which protects the input/ Output storage of DP slave station. DP master station has client data transmission monitoring with viable timer.
Type of device	The second type of DP Master station (DPM2)is a device that can be programmed, configure d and diagnosed. The first type of DP master station (DPM1)is central PLC, e.g. PLC or PC. DP slave station is withthe driver, valve that has binary value or analog in/output

Basic types of PROFIBUS-DP

Each PROFIBUS-DP system can include three different types of the following devices (equipment)

The 1st level DP master station is the central controller, which can exchange information with scattered station (DP slaver station) within the preset period. PLC or PC are typical DPM1.

The 2 level DP master station is a programmer, configuration devices or operation panel, which are used in DP system configuration operation, doing the system operation and monitoring.

DP slave station is the periphery devices which input/output and send out information (I/O device, driver, HMI, and valve).

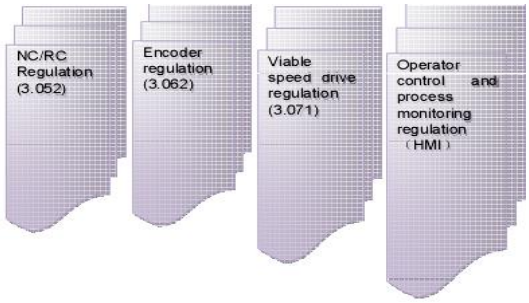
DP slave station is the periphery devices which input/output and send out information (I/O device, driver, HMI, and valve).

Each sub-system includes a DPMI. The Assigned several slave stations and possible

DPM2 device. Any master station can read the input/output image from DP slave station.

There are many master stations connected to the bus. Master station and slave station are independent. But only one DP master station is permitted to write data to DP slave station.

REGULATION OF USING PROFIBUS–DP



Task Two Industrial Control Configuration

In the use of industrial control software, we often mention the word “configuration”. In a word, configuration is a process fulfilling a specific task with the tools and methods provided by the application software.

In the field of industrial control, there is much configuration software, such as

Wincc,inTouch

iFix MCGS

King Configuration

Subtask One Getting to Know Industrial Control Configuration

Application Field of Industrial Control Configuration

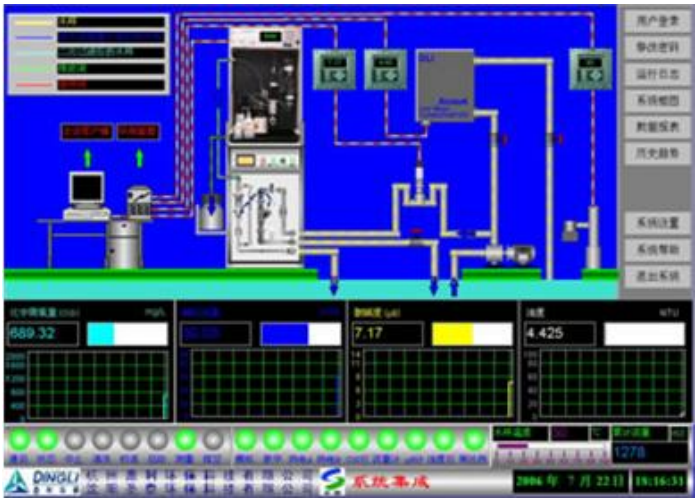
The application of configuration software are very broad, it can be used for data collection, monitoring control, process control, etc. In the filed of power systems, water supply systems, petroleum, and chemical and so on

Subtask Two Getting to Know of MCGS Configuration Software

MCGS (Monitor and Control Generated System), is developed by Beijing Kunluntongtai Automation Software Technology Co., Ltd., which is a set of configuration software system based on Windows platform, used for the rapid configuration and generation of monitoring &control system. It can run on Microsoft Windows2000/Me/NT/operating systems.

Projects Case

Online monitoring system of BMW Brilliance coating production lines is developed by MCGS configuration software, mainly used to help users manage the site online analysis. The software provides the following functions: 1) identity recognition, 2) system flow diagram, 3) dynamic display of monitoring data, 4) operation status of indicating analyzer. 5) set operating parameters of the analyzer, 6) automatically generation of analyzer diary, 7) automatically generation of report forms, 8) inquiry report forms, 9) browsing historical trend curve and all local and remote operation of above features. (IE browser)

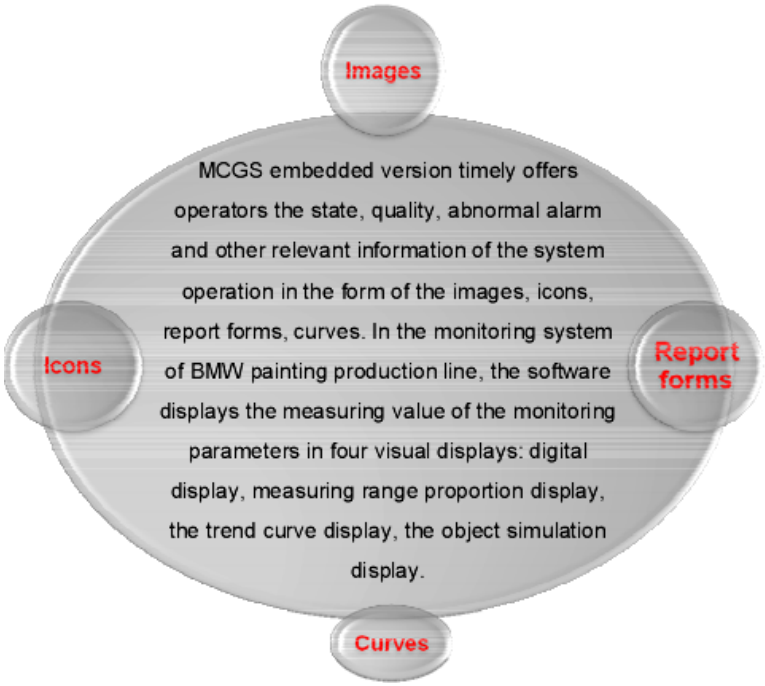


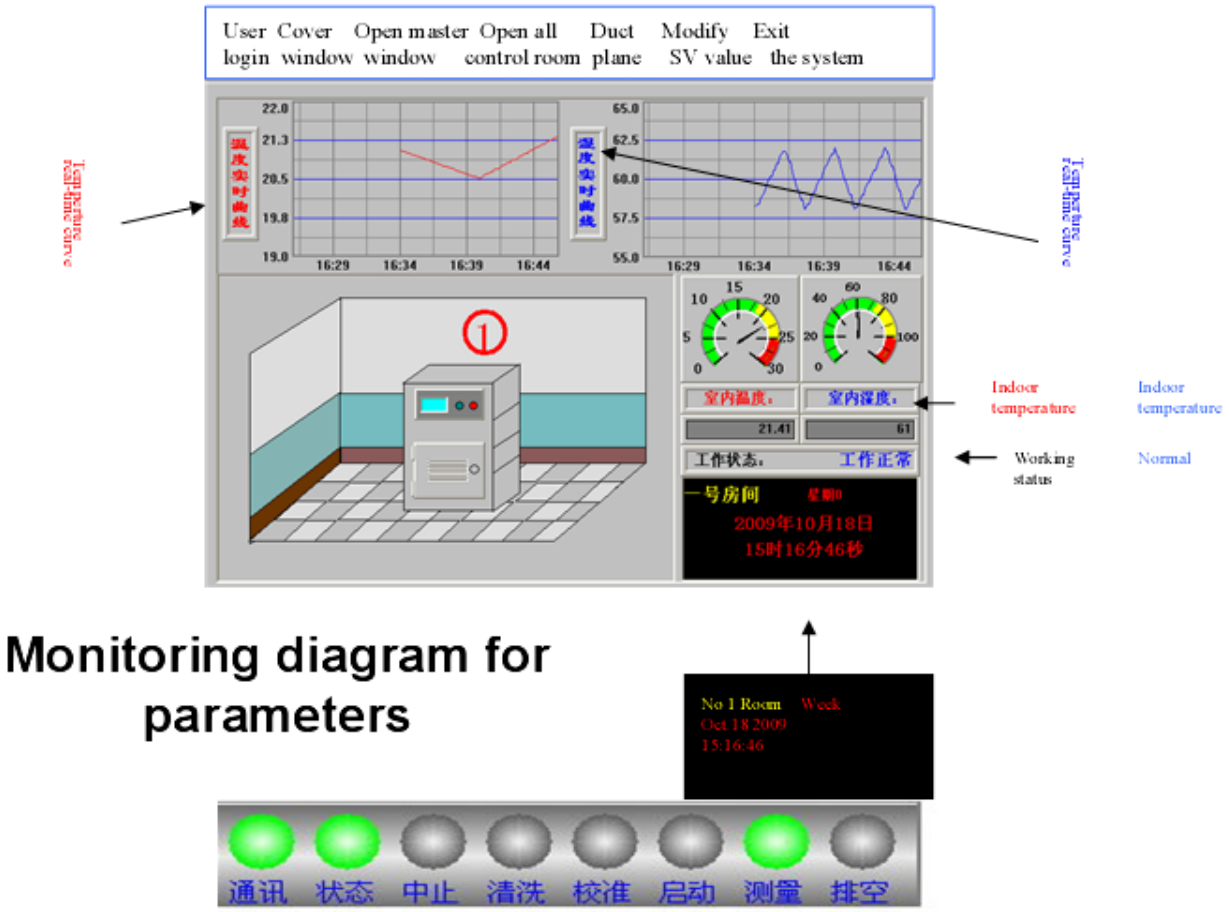
Interface chart of online monitoring system of coating production line in BMW Brilliance

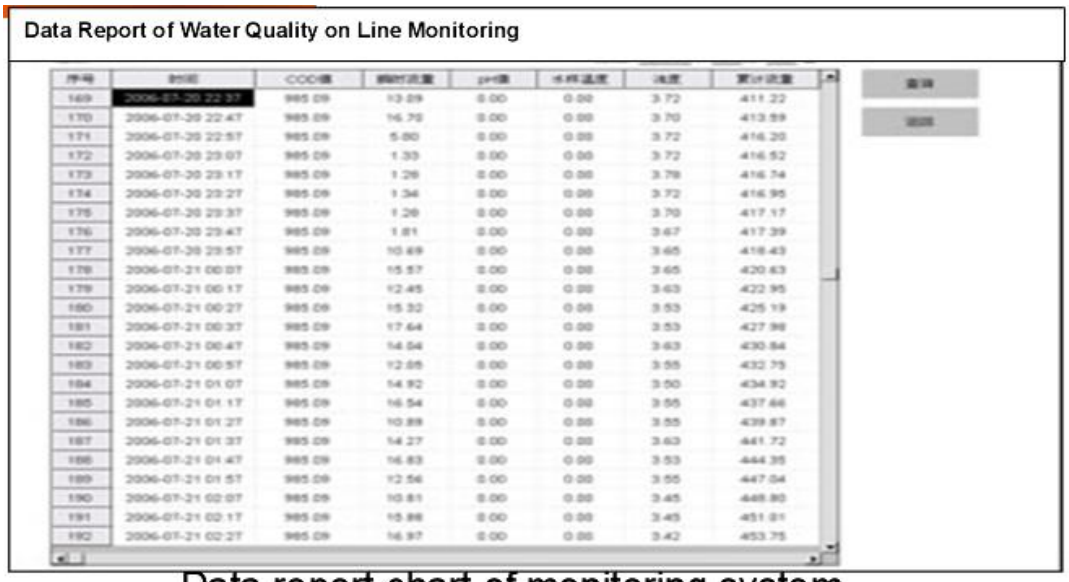
MCGS configuration software can realize the following functions:

1. Dynamic visualization control of the industrial production process;
2. Production data collection and management of the production process;
3. Monitoring and alarming of the production process;
4. Functions of forms for reporting statistics;
5. Web-based data upload and the corresponding control

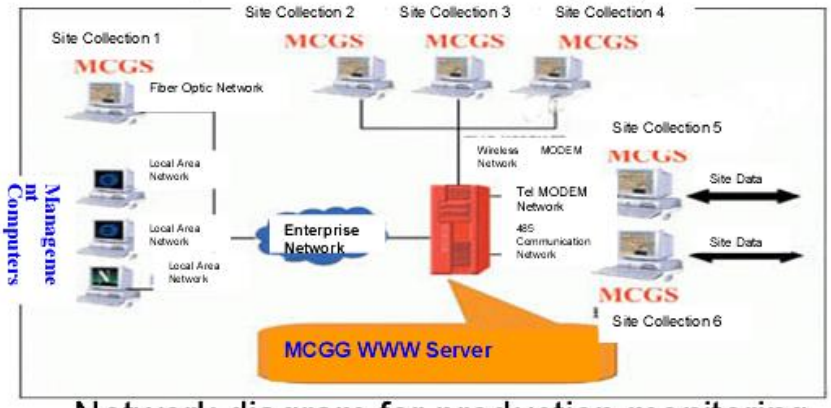
Displays of MCGS embedded version




User login
Cover window
Open master window
Open all control room
Duct plane
Modify SV value
Exit the system
Temperature real-time curve
Temperature real-time curve
Indoor temperature
Indoor temperature
Working status
Normal
Monitoring diagram for parameters
No 1 Room Week
Oct 18 2009
15:16:46
通讯 状态 中止 清洗 校准 启动 测量 排空
Communication Status Stop Rinse Adjustment Start Measuring Empty



Data report chart of monitoring system



Network diagram for production monitoring

Feature of MCGS Embedded Version Configuration Software

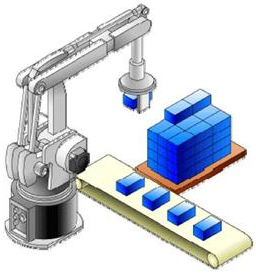
Item	Content
Little capacity	The entire system requires only storage space for minimum configuration. It is easy to use the DOC and other storage devices.
High speed	High precision of the system control time can easily complete a variety of high-speed acquisition system to meet the requirements of real-time control system.
Low cost	Minimum configuration of the system is 386 single board computer with main frequency 24MB, 2MB DOC, 4MB memory. It significantly reduces devices costs.
Real embedded	Running on embedded real-time multi-tasks operating system.
High stability	No hard disk, built-in watchdog, the power-on reset time is short. It can be long-time running constantly in a variety of harsh environments.
Powerful function	Provide interrupt handling, timing scan precision can reach millisecond. Provide the computer serial port, memory, ports access. And offer flexible configuration based on needs.
Convenient communication	Built-in serial communications function, Ethernet communications function, Web browsing function and remote diagnostic functions can easily achieve data exchange, remote collection and Web browsing with a variety of devices(equipment).
Easy to use	The configuration environment of MCGS embedded version, universal version and network version not only inherits their advantage of easy to learn, also adds flexible module operations. It forms the user control system by taking process as a unit. It makes the configuration operation of MCGS embedded version simple, visual and flexible.
Support multiple Devices(equipment)	Provides all commonly used drivers of hardware devices.
Help to set up a complete solution	MCGS embedded version configuration environment runs on a Windows operating system with good man-machine interface. It has the same configuration environment interface as the general and network versions launched by Beijing Kunluntongtai company. It can effectively help users to build complete solutions from embedded devices, on-site monitoring stations to production monitoring information network. It is helpful projects developed by users to move smoothly on these three levels.

Task Three Industrial Robots

Subtask One Getting to Know Industrial Robots

Application of Industrial Robots

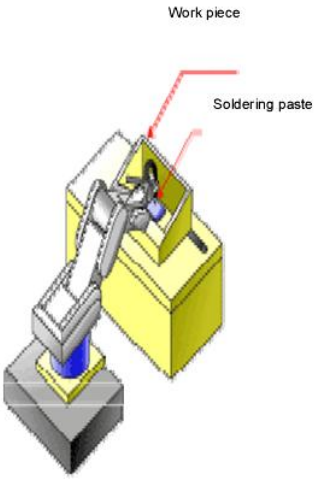
1. Stacking robot



Object operations:

It mainly involves the operations during storing and handling in delivery process and warehouse of the factory.

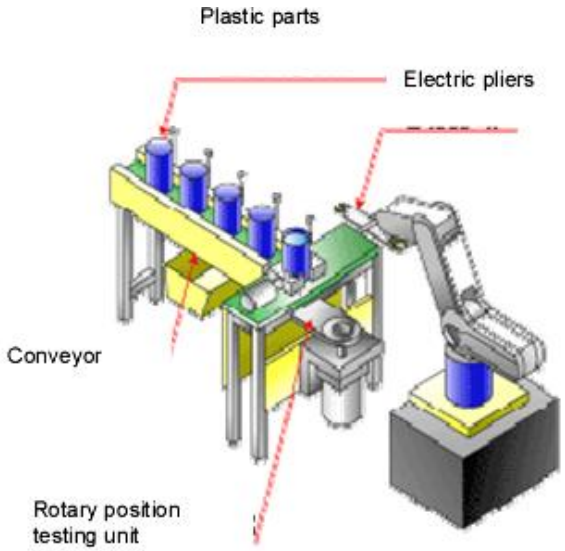
2. Robot for sealing operation



Object operations:

Install an applicator at the front end of the manipulator (mechanical hand), for doing sealants, filler, solder coating.

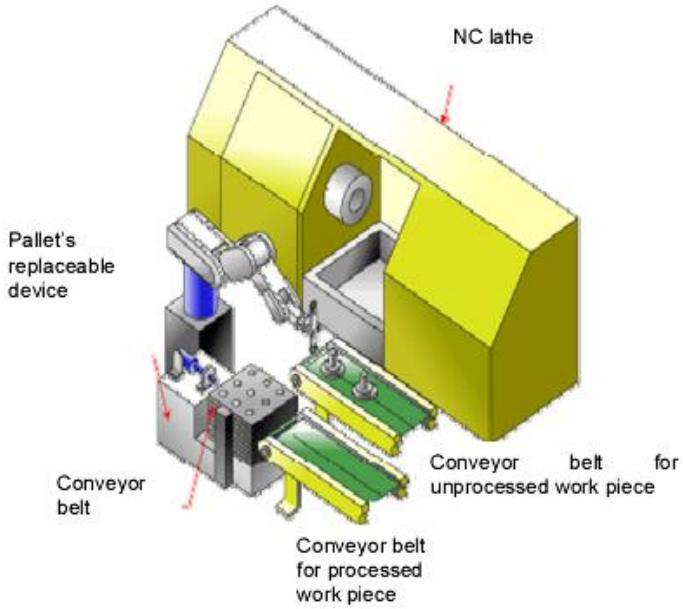
3. Robot for pouring slot cutting



Object operations:

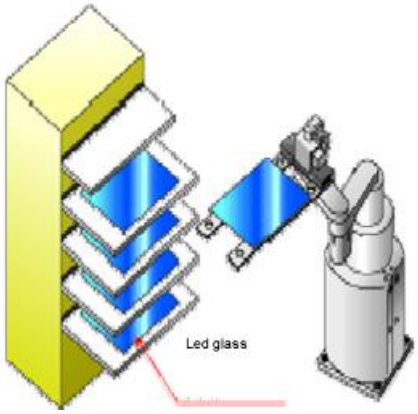
Cut pouring slot generated from plastic injection molding. Install the cutting tools (pliers, etc.) at the front-edge of the manipulator to do the cutting.

4. Robot for loading and unloading work piece



Object operations: load raw work pieces on machine tools (NC lathe) and unload the finished work piece after processing

5. Robot for Cleaning Room



Object Operation:

It is used in semiconductor and LCD manufacturing processes which needs very clean environment

Subtask Two Getting to Know the Properties of Industrial Robots



Manipulator is a multi-functions machine which can automatically position and reprogram. It has multi degrees of freedom to move objects to complete tasks in different environments.

Manipulator consists of the following five components:

Actuator

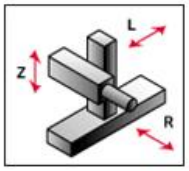
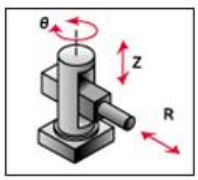
Driver–transmission

Control system

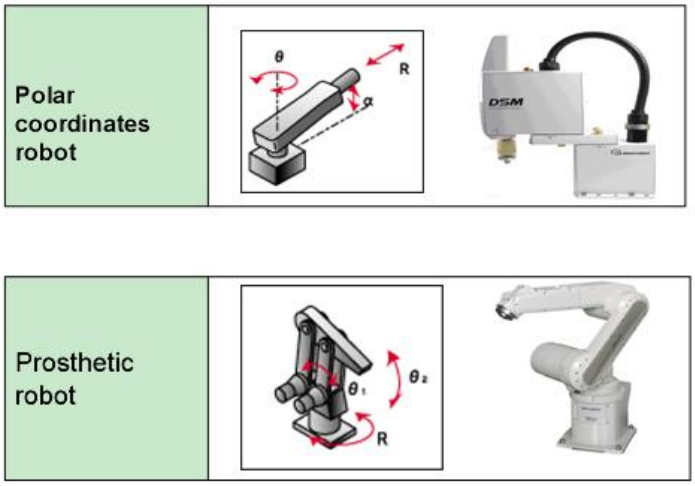
Intelligent system

Remote diagnostic monitoring system

Classifications

Rectangular coordinates robot	 
Cylindrical coordinate robot	 





Task Four Prospects for Flexible Production Line Industrial Control Configuration

Subtask One What is a Flexible Production Line?

Flexible production line, combining the microelectronics with the computer science and systems engineering organically, is a technically complex and highly automated system

COMPOSITION & FUNCTION OF THE FLEXIBLE PRODUCTION LINE

Composition	Function
Automatic processing system	It is a processing system which is based on the group technology and put the work pieces, which is in the similar dimension (not exactly the same shape), about the same weight, material and process, together on the one or several CNC machine or special lathe.
Logistics system	It is composed by a variety of transport equipment (conveyor belts and manipulator) to conduct feeding and delivery of the work piece. It is a main part in flexible production line.
Information system	It collects, handling and feedback of the information required during processing and delivering. It performs multi-level control system to machine tools and delivery equipment by computer or other control devices (hydraulic, pneumatic devices, etc.)
Software system	It is an essential part for effective management to the flexible production line by computer, including design, planning, production control, and system supervision software, etc.

Form	Composition	
Flexible Manufacturing Cell (FMC)	FMC, consisting of 1 or 2 machining centers, has the functions of different cutters' changing and work piece loading & unloading, delivering and storing. Besides the CNC device, there is a cell computer to manage the programs and peripheral devices. FMS fits for the small batch production and the complicated shapes of parts which needs less processing and long time processing.	
Flexible Manufacturing System (FMS)	FMS consists of 2 or more machining centers, cleaning and testing devices (equipment). it is a better delivering and storage system for the cutters and work piece. Apart from scheduling computer, it has process control computers and distributed numerical control terminals to form the local networks with multi-stage control system.	
Autonomy Manufacturing Island (AMI)	The form of AMI is based on group technology. It consists of Based on the several CNC and general-purpose machine tools. group technology. The features of It combines process equipment with production organization, management and manufacturing. It performs the process design, CNC programs management, operation planning and real-time production scheduling. It has a wide range of applications, a low investment and high flexibility.	



Flexible manufacturing cell (FMC)



Flexible manufacturing system (FMS)

MAIN ADVANTAGES OF THE FLEXIBLE PRODUCTION LINE

Advantages	Definition
High utilization	The capacity of one set of machine tool increases several times than that of the single operation after being put in the flexible production line.
High quality	The automatic processing system is composed of one or more machine tools. It Is capable of degradation operation in case of fault. Material delivery system also has the ability to detour the malfunctioned machine tools.
Stable form	The parts are loaded and unloaded once to complete the process during the processing. It has high precision and stable processing form.
Flexible operation	Some of the testing, loading and maintenance can be completed during the first shaft. Unmanned normal production can be achieved during the second and third shaft. In an ideal flexible production line, the monitoring system can also solve some unexpected problems, like the exchange of the worn tools, clearance of the blocked logistics etc.
Great adaptability	Tools, fixtures, and material delivery devices are adjustable. decent Layout of the system is reasonable and suitable for changing the equipments to meet the market needs.

Subtask Two Getting to Know Principle of FPL Process Design

Equipment should be advanced, mature, reliable, stand market test and acquire users' acceptance.

Based on the product accuracy and production program, Meeting needs is enough.

It is fit for the processing, work environment, and habits (operation habits, maintenance habits and equipment selection habits).